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The broken stick problem as the continuum limit of a Bernoulli process

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Abstract

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by

Ulysses Zhan

This research investigates the behavior of the longest success run in a sequence of n Bernoulli trials under a novel continuum limit. In the limit, $n \to \infty$, and the probability $P = p^n$ of having a perfect sequence is kept invariant, where p is the success probability of each Bernoulli trial. We then find the probability distribution of the ratio between the length of the longest run of success and n.

In the continuum limit, the discrete recurrence relation governing the longest run distribution transforms into a continuous integral equation. To resolve the boundary singularities inherent to this system, we introduce a rescaled distribution function and solve the integral equation piece-by-piece across intervals $1/(q+1) < x < 1/q$ using the Adomian decomposition method.

Our analytical solution reveals an exact mathematical equivalence to the classic broken stick problem, where a unit-length interval is partitioned at random points and the distribution of the length of the longest segment is asked. By treating the continuum longest run distribution as a power series in $\lambda = -\ln P$, we demonstrate that the expansion coefficients correspond precisely to the solution of the broken stick problem. These two separate paradigms are thus formally unified through an exponential generating function.

Finally, we show that both the continuum longest run distribution and the broken stick distribution converges to the Gumbel distribution in certain limits. This establishes a rigorous link to the behavior observed in percolation models.

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Chapter 1

Introduction

Runs are an important concept in statistics. A run is defined as a sequence of consecutive identical elements in a random sequence of elements [1, pp. 1–3]. The random sequences where runs are studied the most are Bernoulli processes, where each element is independently and identically distributed with two possible outcomes, conventionally named success (S) and failure (F), with probabilities p and $1 - p$ respectively. This model has significant applications, such as hypothesis testing of whether two distributions are the same [2], assessing the adequacy of curve fitting [3], and start-up demonstration testing [4].

An important problem regarding runs is the probability distribution of the length of the longest run in a finite sequence of n Bernoulli trials. This problem can be converted to the problem of finding the distribution of the waiting time variable corresponding to a run of specific length k (by observing that the probability of the longest run being longer than k in a sequence of n trials is equal to the probability of encountering a run of length k after waiting for at most n trials), the answer to which is called the geometric distribution of order k [5]. Different approaches and equivalent answers arise for this problem [1, pp. 11–17], but one of them is central to the focus of this article, which will

be discussed in Chapter 3.

There are different ways in which an $n \to \infty$ limit can be taken for this problem [1, pp. 29–33]. The behavior of the length of the longest run as $n \to \infty$ while p is kept constant has been studied extensively [6]. It has been shown that the distribution of length of the longest run follows a Gumbel distribution in this limit [7]. This current article focuses on a different $n \to \infty$ limit of the longest run problem, where p is allowed to vary with n such that the probability $P \coloneqq p^n$ of having zero failures is kept constant as $n \to \infty$. This limit is closely related to another classic problem in probability theory, the broken stick problem.

The broken stick problem states as follows [8]. Given a stick of unit length, if it is broken at random k points (independently and uniformly distributed along the stick), it will be divided into $k + 1$ segments. What are the probability distributions of the length of the longest segment, the shortest segment, the second longest segment, etc.? Chapter 10 gives a full solution to the problem. This model has wide applications in various fields, such as ecology [9], machine learning [10], and genomics [11].

Intuitively, if we discretizes the stick as a sequence of n elements, where each element is either intact (S) or broken (F), the longest segment of a broken stick then corresponds to the longest run of S's in the sequence. To this sense, the broken stick problem may be considered the continuum limit of the run statistics problem, where the number of elements $n \to \infty$. In order to keep the number of breaks finite in this limit, we need $P \coloneqq p^n$ to be finite, which is why we consider particularly the limit in which P is held constant as $n \to \infty$.

In this limit, the recurrence relation for the distribution of the longest run derived in Chapter 3 reduces to an integral equation. The equation and its derivation is discussed in Chapter 4. It can be solved using the Adomian decomposition method (ADM) in a piece-by-piece fashion, the precise meaning of which is explained in Chapter 6. Because

the ADM is important for this article, instead of giving an introduction to it here, I will save it in a more complete overview in Chapter 5.

Now, the number of breaks in the stick follows a Poisson distribution with mean $\lambda := -\ln P$ instead of being fixed at k. Therefore, we need to expand the distribution as a power series of λ to get the answers to the broken stick problem as the coefficients. This process is discussed in Chapter 11.

Chapter 2

Notations and conventions

Table 2.1 shows a list of notations used throughout the article. It is not comprehensive in that it does not contain a notation if it is too trivial, only used in a small scope of the article, and does not potentially conflict with notations used elsewhere in the article.

Table 2.1: Glossary of notations used in this article. Sometimes different concepts use the same notation.

Notation	Meaning
i, j, k, l, m, q, s	Summation variables
t, u, v, w	Integration variables
I_1, I_2, I_3	Abbreviations for integrals as they are being calculated
n	Length of the Bernoulli sequence
S, F	Possible outcomes of a Bernoulli trial: sucess, failure
p	Probability of being S for each Bernoulli trial
$P_{n,k}(p)$	Probability that the length of the longest run of S is k
k	Length of the longest run of S
P	Probability that all Bernoulli trials are S

$f(P,x)$	Continuum limit of $P_{n,k}(p)$ (Equation 4.1)
x	Normalized length of the longest run of S
$\delta(\cdot)$	Dirac δ function
$g(P,x)$	Rescaled and finite version of $f(P,x)$ (Equation 4.6)
λ	Order marker for the ADM; equals 1
$g^{(q)}(P,x)$	$g(P,x)$ restricted to domain $1/(q+1) < x < 1/q$
q, s	Order of a piece (as in $1/(q+1) < x < 1/q$ etc.)
$g_j^{(q)}(P,x)$	The jth term when solving $g^{(q)}(P,x)$ using the ADM
r_s	Abbreviation of $\ln P^{sx-1}$
$\Delta g_s(P,x)$	Components of $g^{(q)}(P,x)$ (Equation 7.1)
$A_{s,j}$	Undetermined coefficients in $\Delta g_s(P,x)$ (Equation 7.2)
$B_{s,l}$	Undetermined coefficients depending on $A_{s,j}$ (Equation 7.6)
$\lfloor\cdot\rfloor$, $\lceil\cdot\rceil$	Floor function and ceiling function
$\partial^j/\partial x^j$, ∂_x^j	Taking the jth derivative w.r.t. x while keeping P constant
$\partial_x^j \cdot(x_0)$	Abbreviation of $\partial_x^j \cdot(x)\big\vert_{x=x_0}$ when no ambiguity
$\partial^j/\partial r_q^j$	Taking the jth derivative w.r.t. r_q while keeping P constant
$D_{q,j}$	Jump in $\partial_x^j f(P,x)$ at $x = 1/q$
D_q	$D_{q,q-2}$, i.e., the first nonzero $D_{q,j}$
$T_j(z)$	Touchard polynomial [12]
$F(P,x)$	Twice integral of $f(P,x)$ (Equation 9.1)
$\Delta G_s(P,x)$	Satisfies $\Delta g_s(P,x) = \partial_x^2 \Delta G_s(P,x)$
$\operatorname{Ei} z$	Exponential integral function
γ	Euler–Mascheroni constant
$\operatorname{li} z$	Logarithmic integral function
μ_l	The lth moment of probability density $f(P,x)$

μ	Mean value of the distribution with probability density $f(P, x)$
X_l	Random point at which the stick (the unit interval) is broken
k	Number of points at which the stick is broken
S_l	Length of a segment of the broken stick
$S_{(l)}$	Length of the lth (zero-based) shortest segment
I_l	Boolean random variable indicating whether $S_l > x$
$\mathrm{P}[\mathcal{A}]$	Probability that a (random) proposition $\mathcal{A}$ is true
$\mathrm{P}[\mathcal{A} \mid \mathcal{B}]$	Conditional probability of $\mathcal{A}$ given $\mathcal{B}$
$\mathrm{E}[X]$	Expectation value of a (random) expression X
$\bigwedge_l \mathcal{A}_l$, $\bigvee_l \mathcal{A}_l$	Logical conjunction and disjunction of propositions
z^{*k}	Rectified power function: z^k if $z > 0$; 0 if $z \leq 0$.
λ	Abbreviation of $-\ln P$
∂_λ^k	Taking the kth derivative w.r.t. λ while keeping x constant
$\partial^k / \partial r_s^k$	Taking the kth derivative w.r.t. r_s while keeping x constant
$G(y)$	Integrated CDF of the standard Gumbel distribution
$\bar{\mu}$, β	Parameters of the Gumbel distribution

This article uses the convention $0^0 = 1$.

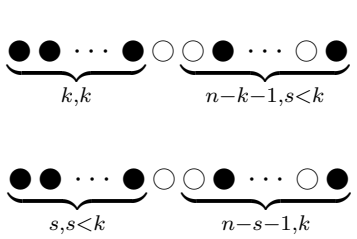
Chapter 3

Recurrence relation for the longest run distribution

The problem of the longest run distribution is formally stated as follows. Consider a Bernoulli sequence of length n, where each element is either success (S) or failure (F), with probabilities p and $1-p$ respectively, and independent of each other. Let $P_{n,k}(p)$ be the probability that the length of the longest run of S's is k. What is the expression of $P_{n,k}(p)$?

Notice that the longest run in a sequence of length n must either start at the beginning of the sequence or not start at the beginning. Therefore, the total probability $P_{n,k}(p)$ for the longest run to be of length k is the sum of the probabilities of these two cases: the probability of having a longest run of length k starting at the beginning, plus the probability of having a longest run of length k not starting at the beginning. An illustration of these two cases is given in Figure 3.1.

The probability that the longest run starts at the beginning and has length k is the product of three probabilities: the probability for the first k elements to be S's, the probability for the next element to be an F, and the probability for the remaining $n-k-1$



$$p^k (1-p) \sum_{s=0}^{k} P_{n-k-1,s}$$

$$\sum_{s=0}^{k-1} p^s (1-p) P_{n-s-1,k}$$

Figure 3.1: The two cases for the longest run of S's to be of length k. Their sum gives the total probability $P_{n,k}(p)$.

elements to have a longest run of length at most k. Therefore, the probability of this case is $p^k (1-p) \sum_{s=0}^{k} P_{n-k-1,s}$ (the explicit dependence on p is suppressed for notational convenience).

If the longest run has length k and does not start at the beginning, this means that the run of S's starting at the beginning of the sequence must be shorter than k. Suppose that the initial run has length $s < k$. Then, the probability of this case is the product of three probabilities: the probability for the first s elements to be S's, the probability for the next element to be an F, and the probability for the remaining $n-s-1$ elements to have a longest run of length k. Summing over all $s < k$ gives the total probability of this case as $\sum_{s=0}^{k-1} p^s (1-p) P_{n-s-1,k}$.

Combining these two cases, we get the recurrence relation

$$P_{n,0<k<n} = (1-p) \left(p^k \sum_{s=0}^{k} P_{n-k-1,s} + \sum_{s=0}^{k-1} p^s P_{n-s-1,k} \right). \tag{3.1}$$

For $k = 0$ and $k \geq n$, the recurrence relation does not hold, but instead we have the trivial cases:

$$P_{n,0} = (1-p)^n, \qquad P_{n,n} = p^n, \qquad P_{n,k>n} = 0. \tag{3.2}$$

Equation 3.1 along with Equation 3.2 completely determines the probability distribution $P_{n,k}(p)$. The expressions $P_{n,k}(p)$ for the first few values of n and k are given in Table 3.1.

It is a polynomial function of p of degree n whose term with the lowest degree is of degree k. The general non-recursive expression of the function is not known in a closed form.

Table 3.1: The expressions $P_{n,k}(p)$ for the first few values of n and k.

$P_{n,k}(p)$	$k=0$	1	2	3	$\cdots$
$n=0$	1				
1	$1-p$	p			
2	$1-2p+p^2$	$2p-2p^2$	p^2		
3	$1-3p+3p^2-p^3$	$3p-5p^2+2p^3$	$2p^2-2p^3$	p^3	
$\vdots$					$\ddots$

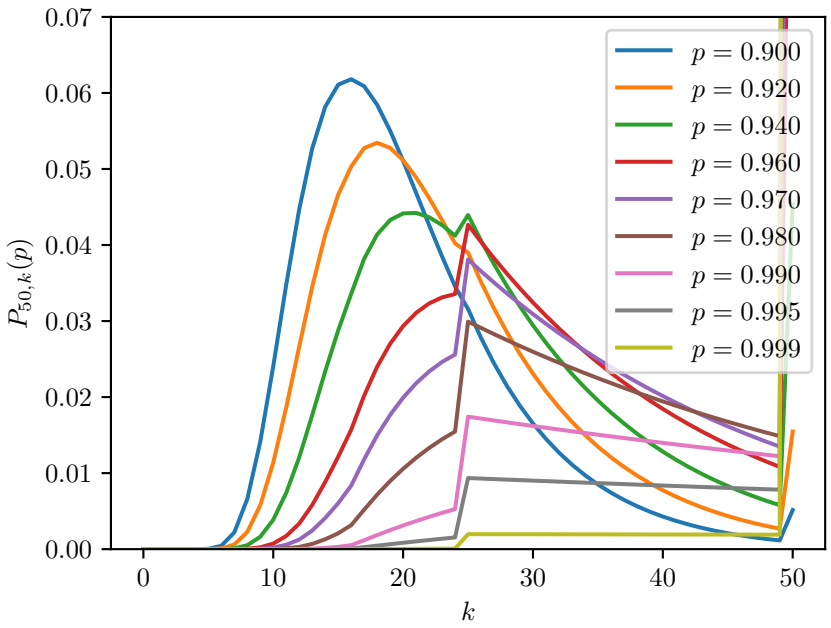


Figure 3.2: The probability distribution $P_{n,k}(p)$ when $n = 50$.

The probability distribution when $n = 50$ is plotted for different values of p in Figure 3.2. Observing the plot, one can see that there is a δ-function-like peak at $k = n$,

a jump in value near $k = n/2$, and a (faintly visible) jump in slope near $k = n/3$. One cannot qualify them as discontinuities because this is a discrete distribution, but such observations are the motivation for studying the continuum limit of the distribution. As we will see later in Chapter 8, there are singularities in the distribution function in the continuum limit that can account for such features that we can see in the discrete distributions.

Chapter 4

Integral equation in the continuum limit

Define $P \coloneqq p^n$ and keep it constant as $n \to \infty$. The probability distribution of $x \coloneqq k/n$ is

$$f(P,x) \coloneqq \lim_{n\to\infty} nP_{n,xn}\left(P^{1/n}\right) \qquad (0 \le P \le 1, \quad x \ge 0)\,. \tag{4.1}$$

This is the continuum limit of the longest run distribution.

The cases when $P = 0$ and $P = 1$ are trivial. We have $f(0,x) = \delta(x)$ and $f(1,x) = \delta(x-1)$. In the rest of this section, we will then assume $0 < P < 1$.

The cases where $x = 0$ or $x = 1$ are covered by the boundary conditions in Equation 3.2. Sbstitute it into Equation 4.1, and we have

$$f(P,0) = \lim_{n\to\infty} n\left(1 - P^{1/n}\right)^n = 0, \tag{4.2}$$

$$f(P,1) = \lim_{n\to\infty} nP = \infty.$$

The infinity at $x = 1$ signifies a δ function peak. From the original statistical problem, we can easily see that the δ function peak at $x = 1$ is due to the finite probablility of

having all n trials in the Bernoulli sequence be S, which is $P = p^n$. Therefore,

$$f(P, x) = \text{finite} + P\delta(x - 1)\,. \tag{4.3}$$

When $0 < x < 1$, substitute Equation 3.1 into Equation 4.1:

$$f(P,x) = \lim_{n\to\infty} n\left(1 - P^{1/n}\right)\left(P^x \sum_{s=0}^{xn} P_{n-xn-1,s}\left(P^{1/n}\right) + \sum_{s=0}^{xn-1} P^{s/n} P_{n-s-1,xn}\left(P^{1/n}\right)\right).$$

Express the limit of a product as the product of limits. In the continuum limit, the summations will become integrations, so we are motivated to sum over a continuous variable t instead of s here. Rewrite:

$$\begin{aligned} f(P,x) = &\lim_{n\to\infty} n\left(1 - P^{1/n}\right) \\ &\cdot \lim_{n\to\infty}\left(P^x \sum_{\substack{t=0 \\ \Delta t = 1/n}}^{x} P_{(1-x)n,tn}\left(P^{1/n}\right) + \sum_{\substack{t=0 \\ \Delta t = 1/n}}^{x} P^t P_{(1-t)n,xn}\left(P^{1/n}\right)\right). \end{aligned}$$

The first limit is simply $-\ln P$. In the second limit, use Equation 4.1 to replace the summands with the continuous distribution. We then have

$$\begin{aligned} f(P,x) = -\ln P \lim_{n\to\infty} \sum_{\substack{t=0 \\ \Delta t = 1/n}}^{x} &\left(P^x \frac{1}{(1-x)\,n} f\left(P^{1-x}, \frac{t}{1-x}\right)\right. \\ &\left. + P^t \frac{1}{(1-t)\,n} f\left(P^{1-t}, \frac{x}{1-t}\right)\right). \end{aligned}$$

Replace the summation with an integration using the definition of the Riemann integral, and we have

$$f(P,x) = -\ln P \int_0^x \mathrm{d}t \left(\frac{P^x}{1-x} f\left(P^{1-x}, \frac{t}{1-x}\right) + \frac{P^x}{1-t} f\left(P^{1-t}, \frac{x}{1-t}\right)\right).$$

Notice that this equation naturally satisfies Equation 4.2. Therefore, to have an equation that includes all values in $0 \le x \le 1$, we just need to incorporate the δ function

peak in Equation 4.3. For x outside the interval, $f(P,x)$ vanishes. This gives us

$$\begin{aligned} f(P,0\le x\le 1) = & -\ln P\int_0^x \mathrm{d}t\left(\frac{P^x}{1-x}f\left(P^{1-x},\frac{t}{1-x}\right)\right. \\ & \left.+\frac{P^x}{1-t}f\left(P^{1-t},\frac{x}{1-t}\right)\right)+P\delta(x-1)\,, \end{aligned} \tag{4.4}$$

$$f(P,x>1)=0.$$

Together with the normalization condition

$$\int_0^\infty f(P,x)\,\mathrm{d}x = 1, \tag{4.5}$$

we can in principle solve f from these equations.

The presence of the δ function term causes technical difficulties in dealing with the equations, but we can easily remove it by defining a new function

$$g(P,x) \coloneqq \frac{1}{P}f(P,x)-\delta(x-1) \tag{4.6}$$

which is finite everywhere. Substitute Equation 4.6 into Equation 4.4, and we have the equations for g:

$$\begin{aligned} g(P,0\le x\le 1) = & -\ln P\left(\int_0^{x/(1-x)}\left(g\left(P^{1-x},u\right)+\delta(u-1)\right)\mathrm{d}u\right. \\ & \left.+\int_x^{x/(1-x)}\left(g\left(P^{x/v},v\right)+\delta(v-1)\right)\frac{\mathrm{d}v}{v}\right), \end{aligned} \tag{4.7}$$

$$g(P,x>1)=0.$$

Here, the integration variables are changed to $u \coloneqq t/(1-x)$ and $v \coloneqq x/(1-t)$ for technical convenience. The normalization condition for g is obtained by integrating Equation 4.6 on both sides and substitute Equation 4.5:

$$\int_0^\infty g(P,x)\,\mathrm{d}x = \frac{1}{P}-1. \tag{4.8}$$

These equations can be solved piece-by-piece using the Adomian decomposition method as will be illustrated in Chapter 6. After obtaining the solution for g, we can recover f

from g using

$$f(P,x) = P\left(g(P,x) + \delta(x-1)\right).$$

Chapter 5

Overview of the Adomian decomposition method

The Adomian decomposition method (ADM) is a method for solving ordinary and partial differential equations, integral equations, and other functional equations [13]. The core idea of the ADM is to decompose the unknown function as the sum of an infinite series, and then iteratively compute the terms of the series by comparing the terms on both sides of the equation.

Generally, the first step of the ADM is to rewrite the equation we want to solve in the form

$$g = \Phi(g)\,, \tag{5.1}$$

where g is the unknown function, and Φ is an operator that maps a function to a function. Assume that Φ is sufficiently smooth so that it makes sense to expand it in a Volterra series [14] (see e.g. [15] for an example of this type of expansion)

$$\Phi(g) = \Phi_0 + \lambda\Phi_1(g) + \lambda^2\Phi_2(g,g) + \lambda^3\Phi_3(g,g,g) + \cdots\,,$$

where $\lambda = 1$ is used to label the order, and Φ_j is a symmetric j-linear map.

Next, we express g as an infinite series

$$g = g_0 + \lambda g_1 + \lambda^2 g_2 + \lambda^3 g_3 + \cdots ,$$

where $\lambda = 1$ is again used to label the order, and g_j are all unknown functions that are to be solved iteratively. Assuming that Φ_j is continuous in the topology where the series expansion of g converges, we can expand both sides of Equation 5.1 as series in λ:

$$g_0 + \lambda g_1 + \lambda^2 g_2 + \cdots = \Phi_0 + \lambda \Phi_1(g_0) + \lambda^2 \left(\Phi_1(g_1) + \Phi_2(g_0, g_0)\right) + \cdots$$

Then, we can compare the λ^j term on both sides to get an expression for g_j. Generally, we have [13, p. 11]

$$g_j = \sum_{l=0}^{j} \sum_{m_1 + \cdots + m_l = j-l} \Phi_l(g_{m_1}, \ldots, g_{m_l}) \,. \tag{5.2}$$

This recurrence relation can be used to find all g_j iteratively, and then one can sum all of them to find the solution g.

When employing this method, one needs to be careful to check that the series indeed converges and that the resultant sum is indeed a solution to the original equation. This is because the whole method relies on the assumption that such a convergent series expansion exists, and some steps in the series expansions require non-trivial smoothness or continuity conditions. For example, if $\Phi(g)$ is an integration involving g (such as a convolution), then expanding it as a series of λ involves interchanging summation and integration, which is not always permitted (a sufficient condition is the absolute convergence by Fubini's theorem).

Specially, when using the ADM to solve a (nonhomogeneous) linear equation, we have $\Phi_j = 0$ for all $j > 1$. Then, from Equation 5.2, we have in this case

$$g_j = \Phi_1^{(j)}(\Phi_0) \,, \tag{5.3}$$

where $\Phi_1^{(j)}$ is the jth self-composition of Φ_1.

Chapter 6

Solving the equation using the ADM

The ADM can be used to solve some integral equations, notably Volterra equations [13, p. 224], but the same method can be applied to much more general integral equations, including those like Equation 4.7. In this section, we solve Equation 4.7 and Equation 4.8 using the ADM.

Notice that Equation 4.7 is already of the form of Equation 5.1. However, we cannot expand it right away into a series of Φ_j. Instead, we will solve the equation piece by piece, and the parts that go into Φ_0 and Φ_1 are different for each piece. Inspired by the piecewise behavior shown in Figure 3.2, each piece is the interval $1/(q+1) < x < 1/q$ for $q = 1, 2, \ldots$, and we use a different notation for g on that interval:

$$g^{(q)}(P, x) \coloneqq g(P, x) \qquad \left(\frac{1}{q+1} < x < \frac{1}{q}\right).$$

Notice that the only difference between g and $g^{(q)}$ is the domain on which the function is defined. We will call q the order of the function $g^{(q)}$ and the order of the interval $1/q < x < 1/(q+1)$. We can call $g^{(q)}$ the piece of g with order q. In the rest of this section, we will first use the ADM to find the closed forms of $g^{(1)}$ (Section 6.1), $g^{(2)}$ (Section 6.2), and $g^{(3)}$ (Section 6.3) one by one. Then, we will guess a general expression

for $g^{(q)}$ with undetermined coefficients and find the coefficients in Chapter 7.

An illustration of the process of solving g piece by piece is shown in Figure 6.1. The three horizontal lines from bottom to top for each q represent, respectively, the range of x, the largest interval of integration over u, and the largest interval of integration over v, as they appear in Equation 4.7 when $g^{(q)}$ is being solved. Every time we go up to a higher order, the solution in the last order is utilized to find parts of the integrals. In this way, the arguments of g in each of its appearances in the equation can be limited to be in the domain of $g^{(q)}$ so that we can use the ADM to solve $g^{(q)}$. The process will be clear as we go through it in the rest of the section.

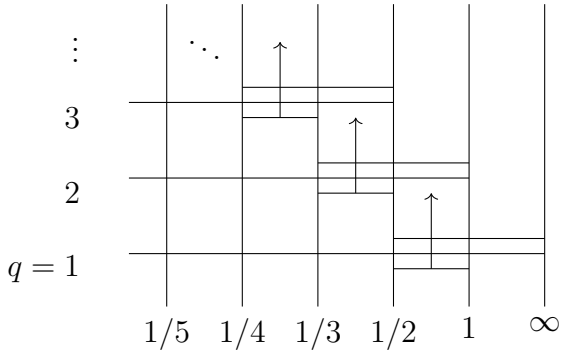


Figure 6.1: Illustration of solving the function g piece by piece using the ADM. The three horizontal lines from bottom to top for each q represent, respectively, the range of x, the largest interval of integration over u, and the largest interval of integration over v, as they appear in Equation 4.7 when $g^{(q)}$ is being solved.

6.1 The first piece

In this subsection, we will look at $g^{(1)}$, which is defined on the interval $1/2 < x < 1$. In this case, we have $x/(1-x) > 1$, so both δ functions in Equation 4.7 are picked up by the integral. In the integral over u, the first term in the integrand integrates exactly

to the normalization condition in Equation 4.8. Therefore, we have

$$g^{(1)}(P,x) = -\ln P \left(P^{x-1} + 1 + \int_x^1 g^{(1)}\left(P^{x/v}, v\right) \frac{\mathrm{d}v}{v} \right). \tag{6.1}$$

Notice that the upper limit of integration can be clamped to 1 because $g(P, x > 1) = 0$. We now apply the ADM to solve this equation. The right-hand side is clearly linear to $g^{(1)}$ plus a constant function. Therefore, the Volterra series only consists of two nonzero terms:

$$\Phi_0(P,x) = -\ln P \left(P^{x-1} + 1\right),$$
$$\Phi_1\left(g^{(1)}\right)(P,x) = -\ln P \int_x^1 g^{(1)}\left(P^{x/v}, v\right) \frac{\mathrm{d}v}{v}.$$

All Φ_j for $j > 1$ vanish. Then, according to Equation 5.3, we have

$$g_0^{(1)} = -\ln P \left(P^{x-1} + 1\right),$$
$$g_{j+1}^{(1)} = -\ln P \int_x^1 g_j^{(1)}\left(P^{x/v}, v\right) \frac{\mathrm{d}v}{v} \qquad (j = 0, 1, \ldots).$$

If the sum

$$g^{(1)} = \sum_{j=0}^{\infty} g_j^{(1)}$$

converges, then this is a guess of the solution to Equation 6.1, which we can verify whether it is correct or not by simply plugging the guess into the equation.

One can find the first few terms of $g_j^{(1)}$. Using those terms, we can guess a general expression, and we do not need to prove the general expression because we will verify anyway whether it yields the correct solution after we sum over j. After doing the labor of calulation (one may find it useful to change the integration variable to $1/v$ when

calculating the integrals), the first few terms are:

$$\begin{aligned}
g_0^{(1)}(P,x) &= -\ln P\left(\mathrm{e}^{r_1}+1\right),\\
g_1^{(1)}(P,x) &= -\ln P\left(\mathrm{e}^{r_1}-1+r_1\right),\\
g_2^{(1)}(P,x) &= -\ln P\left(\mathrm{e}^{r_1}-1-r_1+\frac{1}{2}r_1^2\right),\\
g_3^{(1)}(P,x) &= -\ln P\left(\mathrm{e}^{r_1}-1-r_1-\frac{1}{2}r_1^2+\frac{1}{6}r_1^3\right),\\
&\vdots \quad ,
\end{aligned}$$

where $r_1 \coloneqq \ln P^{x-1}$ is a notation introduced for abbreviation. Clearly seeing the pattern, we guess that the general expression is

$$\frac{g_j^{(1)}(P,x)}{-\ln P} = \mathrm{e}^{r_1} + \frac{1}{j!}r_1^j - \sum_{k=0}^{j-1}\frac{1}{k!}r_1^k.$$

Summing over j is not a trivial calculation, so I will include steps here. Summing over j for the second term is standard; for the other terms, express the infinite sum as the limit for the upper limit of j to tend to infinity:

$$\frac{g^{(1)}(P,x)}{-\ln P} = \sum_{j=0}^{\infty}\frac{g_j^{(1)}(P,x)}{-\ln P} = \mathrm{e}^{r_1} + \lim_{J\to\infty}\sum_{j=0}^{J}\left(\mathrm{e}^{r_1} - \sum_{k=0}^{j-1}\frac{1}{k!}r_1^k\right).$$

Exchanging the summation over k with the summation over j, we have $\sum_{j=0}^{J}\sum_{k=0}^{j-1}$ become $\sum_{k=0}^{J-1}\sum_{j=k+1}^{J}$. Now, $\sum_{j=k+1}^{J} = J-k$. Therefore,

$$\frac{g^{(1)}(P,x)}{-\ln P} = \mathrm{e}^{r_1} + \lim_{J\to\infty}\left((J+1)\,\mathrm{e}^{r_1} - \sum_{k=0}^{J-1}\frac{J-k}{k!}r_1^k\right) = (2+r_1)\,\mathrm{e}^{r_1}. \tag{6.2}$$

One can then plug this result into Equation 6.1 to verify that it is indeed the solution.

6.2 The second piece

In this subsection, we look at $g^{(2)}$, which is defined on $1/3 < x < 1/2$. In this case, we have $1/2 < x/(1-x) < 1$, so neither of the δ functions in Equation 4.7 is picked up by

the integration. The integration over u can again be evaluated using the normalization condition (Equation 4.8), but we also need $g^{(1)}$. To elaborate,

$$\int_0^{x/(1-x)} g\big(P^{1-x}, u\big)\,\mathrm{d}u = \int_0^1 g\big(P^{1-x}, u\big)\,\mathrm{d}u - \int_{x/(1-x)}^1 g^{(1)}\big(P^{1-x}, u\big)\,.$$

The first term can be directly derived from Equation 4.8, and one can calculate the second term using the solution to $g^{(1)}$ that we already have (Equation 6.2). The integration over v can also be expressed so that it only involves $g^{(2)}$ by utilizing the solution to $g^{(1)}$. To elaborate,

$$\int_x^{x/(1-x)} g\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v} = \int_x^{1/2} g^{(2)}\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v} + \int_{1/2}^{x/(1-x)} g^{(1)}\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v}.$$

One can then calculate the second term using the solution to $g^{(1)}$. In the end, Equation 4.7 becomes

$$\frac{g^{(2)}(P,x)}{-\ln P} = P^{x-1} + P^{-x}\left(1+\ln P^{-x}\right) - 2P^{2x-1}\left(1+\ln P^{2x-1}\right) + \int_x^{1/2} g^{(2)}\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v}. \tag{6.3}$$

We can again apply the ADM to solve this equation. The right-hand side is again linear to $g^{(2)}$ plus a constant function, so we can express the solution to $g^{(2)}$ as a series

$$g^{(2)} = \sum_{j=0}^{\infty} g_j^{(2)},$$

whose terms are given by Equation 5.3:

$$\begin{aligned} g_0^{(2)}(P,x) &= -\ln P\left(P^{x-1} + P^{-x}\left(1+\ln P^{-x}\right) - 2P^{2x-1}\left(1+\ln P^{2x-1}\right)\right), \\ g_{j+1}^{(2)}(P,x) &= -\ln P \int_x^{1/2} g_j^{(2)}\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v} \qquad (j=0,1,\ldots)\,. \end{aligned}$$

The same procedure for deriving $g^{(1)}$ can be used here. We first calculate the first few terms in the series, and then we guess the general expression for the terms, and finally we evaluate the sum to get the solution, which we can plug into Equation 6.3 to verify

its correctness. I am not including the first few terms in this article because they go too long to be written here before one may find the pattern. It is advised that one use a CAS software to calculate them, and after getting about six terms can they find the pattern. The general expression is

$$\frac{g_j^{(2)}(P,x)}{-\ln P} = \mathrm{e}^{r_1} + 2\mathrm{e}^{r_2}\,(j-1-r_2) - 2\sum_{k=0}^{j-1}\frac{j-k-1}{k!}r_2^j \\ - P^{-x}\sum_{k=0}^{j-1}\frac{1}{k!}r_2^j + \frac{1}{j!}P^{-x}\left(1+\ln P^{-x}\right)r_2^j,$$

where $r_s \coloneqq \ln P^{sx-1}$ is a notation introduced for abbreviation. Summing over j is also a tedious calculation, and one may find that

$$\frac{g^{(2)}(P,x)}{-\ln P} = (2+r_1)\,\mathrm{e}^{r_1} - \left(2+4r_2+r_2^2\right)\mathrm{e}^{r_2}. \tag{6.4}$$

By plugging this into Equation 6.3, one can verify that it is indeed the solution.

6.3 The third piece

The method for getting $g^{(3)}$ is very similar to the method for getting $g^{(2)}$. For the case of $g^{(3)}$, we have $1/4 < x < 1/3$ and thus $1/3 < x/(1-x) < 1/2$. By using this fact and the solutions of $g^{(1)}$ and $g^{(2)}$ that we already obtain, we can reduce Equation 4.7 into a form such that $g^{(3)}$ equals a linear operator acting on $g^{(3)}$ plus some constant function. One can then use the ADM to solve this system. Because the calculation gets very tedious and because the intermediate results get very long, I only include the final result for $g^{(3)}$ here:

$$\frac{g^{(3)}(P,x)}{-\ln P} = (2+r_1)\,\mathrm{e}^{r_1} - \left(2+4r_2+r_2^2\right)\mathrm{e}^{r_2} + \left(3r_3+3r_3^2+\frac{1}{2}r_3^3\right)\mathrm{e}^{r_3}, \tag{6.5}$$

where $r_s \coloneqq \ln P^{sx-1}$ is introduced for abbreviation.

Chapter 7

Full solution

With the results for $g^{(1)}$, $g^{(2)}$, and $g^{(3)}$, one may guess the form of higher order solutions with undetermined coefficients. A reasonable guess is

$$g^{(q)}(P,x) = \sum_{s=1}^{q} \Delta g_s(P,x)\,, \tag{7.1}$$

where

$$\Delta g_s(P,x) \coloneqq (-1)^s\, \mathrm{e}^{r_s} \ln P \sum_{j=0}^{s} \frac{A_{s,j}}{j!} r_s^j, \tag{7.2}$$

where $A_{s,j}$ are coefficients to be determined. We can then substitute this guessed form into Equation 4.7. To avoid considering the δ functions, we only consider $q > 1$, in which case $x/(1-x) < 1/(q-1) \leq 1$.

Solving the undetermined coefficients consists of mainly two steps: first, expand both sides of Equation 4.7 in terms of those coefficients by calculating the integrals (Section 7.1), and second, compare corresponding terms on both sides to solve the coefficients (Section 7.2).

7.1 Calculating integrals

For the integral over u, express it as the full integral (as in Equation 4.8) minus excluded pieces with different orders. An illustration of this is shown in Figure 7.1, where the dashed line is excluded pieces.

$$\int_0^{x/(1-x)} = \int_0^1 - \sum_{m=1}^{q-2} \int_{1/(m+1)}^{1/m} - \int_{x/1-x}^{1/(q-1)} .$$

The first term can be directly derived using Equation 4.8. The integrands of the second term and the third term are respectively $g^{(m)}(P^{1-x}, u)$ and $g^{(q-1)}(P^{1-x}, u)$, both of which can be expressed as a sum over Δg_s using Equation 7.2. Therefore,

$$\int_0^{x/(1-x)} g\big(P^{1-x}, u\big)\,\mathrm{d}u = P^{x-1} - 1 - \left(\sum_{m=1}^{q-2} \int_{1/(m+1)}^{1/m} \sum_{s=1}^{m} + \int_{x/(1-x)}^{1/(q-1)} \sum_{s=1}^{q-1} \right) \Delta g_s\big(P^{1-x}, u\big)\,\mathrm{d}u.$$

Change the order of summation and integration, and we have

$$\int_0^{x/(1-x)} g\big(P^{1-x}, u\big)\,\mathrm{d}u = P^{x-1} - 1 - \sum_{s=1}^{q-1} \left(\sum_{m=s}^{q-2} \int_{1/(m+1)}^{1/m} + \int_{x/(1-x)}^{1/(q-1)} \right) \Delta g_s\big(P^{1-x}, u\big)\,\mathrm{d}u.$$

Notice that I took the freedom of increasing the upper limit of s in the first term in the parentheses to $q-1$ because the sum over m is zero when $s > q-2$ after exchanging the order of summation anyway. Now, the integrations in the parentheses combine to be a full integration from $x/(1-x)$ to $1/s$, so

$$\int_0^{x/(1-x)} g\big(P^{1-x}, u\big)\,\mathrm{d}u = P^{x-1} - 1 - \sum_{s=1}^{q-1} \int_{x/(1-x)}^{1/s} \Delta g_s\big(P^{1-x}, u\big)\,\mathrm{d}u. \tag{7.3}$$

Denote the integral in Equation 7.3 as I_1. Using Equation 7.2, we have

$$\begin{aligned} I_1 &\coloneqq \int_{x/(1-x)}^{1/s} \Delta g_s\big(P^{1-x}, u\big)\,\mathrm{d}u \\ &= \int_{x/(1-x)}^{1/s} (-1)^s\, P^{(su-1)(1-x)} \ln P^{1-x} \sum_{j=0}^{s} \frac{A_{s,j}}{j!} \left(\ln P^{(su-1)(1-x)}\right)^j \mathrm{d}u. \end{aligned}$$

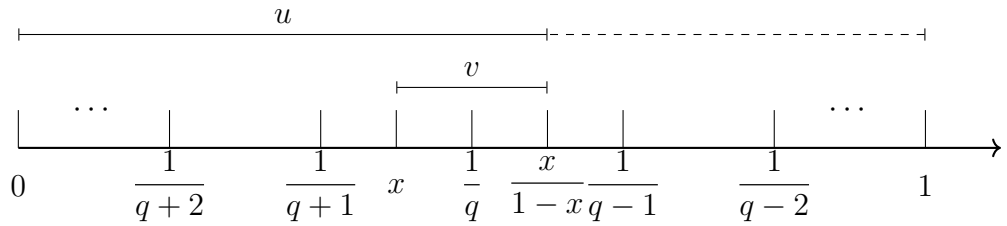


Figure 7.1: Illustration of the interval of integration over u and v when $1/(q+1) < x < 1/q$.

Change the integration variable to $w \coloneqq P^{(su-1)(1-x)}$, and we have

$$I_1 = \frac{(-1)^s}{s} \sum_{j=0}^{s} \frac{A_{s,j}}{j!} \int_{P^{(s+1)x-1}}^{1} (\ln w)^j \, \mathrm{d}w.$$

To carry out the integral, one may observe a handy integral formula:

$$\int (\ln w)^j \, \mathrm{d}w = (-1)^j \, j! \, w \sum_{l=0}^{j} (-1)^l \, \frac{(\ln w)^l}{l!} + C. \tag{7.4}$$

The formula can be proven using mathematical induction and integration by parts. After some calculation, we then have

$$I_1 = \frac{(-1)^s}{s} \left(B_{s,0} - \mathrm{e}^{r_{s+1}} \sum_{l=0}^{s} \frac{B_{s,l}}{l!} r_{s+1}^l \right), \tag{7.5}$$

where

$$B_{s,l} \coloneqq (-1)^l \sum_{j=l}^{s} (-1)^j \, A_{s,j}. \tag{7.6}$$

For the integration over v, split it into pieces with different orders. The split is also shown in Figure 7.1.

$$\int_{x}^{x/(1-x)} = \int_{x}^{1/q} + \int_{1/q}^{x/(1-x)}.$$

The integrands of the two terms are respectively $g^{(q)}(P^{1-x}, u)$ and $g^{(q-1)}(P^{1-x}, u)$. They can be expressed as sums over Δg_s using Equation 7.2. We then have

$$\int_{x}^{x/(1-x)} g\big(P^{x/v}, v\big) \, \frac{\mathrm{d}v}{v} = \left(\int_{x}^{1/q} \sum_{s=1}^{q} + \int_{1/q}^{x/(1-x)} \sum_{s=1}^{q-1} \right) \Delta g_s\big(P^{x/v}, v\big) \, \frac{\mathrm{d}v}{v}.$$

The sums over s up to $q-1$ in the two terms can be combined, and there is a leftover for $s = q$ in the first term, so

$$\int_x^{x/(1-x)} g\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v} = \sum_{s=1}^{q-1} \int_x^{x/(1-x)} \Delta g_s\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v} + \int_x^{1/q} \Delta g_q\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v}. \quad (7.7)$$

Denote the two integrals in Equation 7.7 as I_2 and I_3 respectively. From Equation 7.2, we get

$$\begin{aligned} I_2 &\coloneqq \int_x^{x/(1-x)} \Delta g_s\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v} \\ &= \int_x^{x/(1-x)} (-1)^s\, P^{(sv-1)x/v} \ln P^{x/v} \sum_{j=0}^{s} \frac{A_{s,j}}{j!} \left(\ln P^{(sv-1)x/v}\right)^j \frac{\mathrm{d}v}{v}. \end{aligned}$$

Change the integration variable to $w \coloneqq P^{(sv-1)x/v}$, and we have

$$I_2 = (-1)^s \sum_{j=0}^{s} \frac{A_{s,j}}{j!} \int_{P^{sx-1}}^{P^{(s+1)x-1}} (\ln w)^j\,\mathrm{d}w.$$

Use Equation 7.4, and we have

$$I_2 = (-1)^s \left(\mathrm{e}^{r_{s+1}} \sum_{l=0}^{s} \frac{B_{s,l}}{l!} r_{s+1}^l - \mathrm{e}^{r_s} \sum_{l=0}^{s} \frac{B_{s,l}}{l!} r_s^l \right), \quad (7.8)$$

where $B_{s,l}$ is defined in Equation 7.6. For I_3, we have from Equation 7.2 that

$$\begin{aligned} I_3 &\coloneqq \int_x^{1/q} \Delta g_q\big(P^{x/v}, v\big)\,\frac{\mathrm{d}v}{v} \\ &= \int_x^{1/q} (-1)^q\, P^{(qv-1)x/v} \ln P^{x/v} \sum_{j=0}^{q} \frac{A_{q,j}}{j!} \left(\ln P^{(qv-1)x/v}\right)^j \frac{\mathrm{d}v}{v}. \end{aligned}$$

The calculation is the almost the same as that for I_2. In the end, the result is

$$I_3 = (-1)^q \left(B_{q,0} - \mathrm{e}^{r_q} \sum_{l=0}^{q} \frac{B_{q,l}}{l!} r_q^l \right). \quad (7.9)$$

Now, substitute everything back: substitute Equation 7.2 into the left-hand side of Equation 4.7; substitute Equation 7.3 and Equation 7.7 into the right-hand side of Equation 4.7. The result is

$$\sum_{s=1}^{q} (-1)^s\, \mathrm{e}^{r_s} \ln P \sum_{j=0}^{s} \frac{A_{s,j}}{j!} r_s^j = -\ln P \left(\mathrm{e}^{r_1} - 1 - \sum_{s=1}^{q-1} I_1 + \sum_{s=1}^{q-1} I_2 + I_3 \right).$$

Cancel a factor of $\ln P$ on both sides, and substitute Equation 7.5, Equation 7.8, and Equation 7.9, and we have

$$\begin{aligned}
&\sum_{s=1}^{q} (-1)^s \, \mathrm{e}^{r_s} \sum_{j=0}^{s} \frac{A_{s,j}}{j!} r_s^j \\
= \; & -\mathrm{e}^{r_1} + 1 + \sum_{s=1}^{q-1} \frac{(-1)^s}{s} \left(B_{s,0} - \mathrm{e}^{r_{s+1}} \sum_{l=0}^{s} \frac{B_{s,l}}{l!} r_{s+1}^l \right) \\
& - \sum_{s=1}^{q-1} (-1)^s \left(\mathrm{e}^{r_{s+1}} \sum_{l=0}^{s} \frac{B_{s,l}}{l!} r_{s+1}^l - \mathrm{e}^{r_s} \sum_{l=0}^{s} \frac{B_{s,l}}{l!} r_s^l \right) - (-1)^q \left(B_{q,0} - \mathrm{e}^{r_q} \sum_{l=0}^{q} \frac{B_{q,l}}{l!} r_q^l \right).
\end{aligned} \tag{7.10}$$

The work left to do is now to compare the coefficient of the $\mathrm{e}^{r_s} r_s^j$ term on both sides for every (s, j).

7.2 Solving undetermined coefficients

Compare the $\mathrm{e}^{r_0} r_0^0$ term (the constant term) in both sides of Equation 7.10, and we have

$$0 = 1 + \sum_{s=1}^{q-1} \frac{(-1)^s}{s} B_{s,0} - (-1)^q \, B_{q,0}. \tag{7.11}$$

This is a recurrence relation that can be used to find every term in the sequence $\{B_{q,0}\}$ given one term. Substitute $q \to q+1$ in Equation 7.11 and subtract it from Equation 7.11, and we have

$$B_{q+1,0} = -\frac{q+1}{q} B_{q,0} \qquad (q = 2, 3, \ldots), \tag{7.12}$$

which is a more friendly recurrence relation. The relation between $B_{2,0}$ and $B_{1,0}$, however, cannot be given by Equation 7.12, but it can be derived from putting $q = 2$ in Equation 7.11:

$$B_{2,0} = 1 - B_{1,0}. \tag{7.13}$$

Comparing the $\mathrm{e}^{r_1} r_1^1$ term in both sides of Equation 7.10 gives $A_{1,1} = B_{1,1}$, which is tautology according to Equation 7.6. Comparing the $\mathrm{e}^{r_1} r_1^0$ term gives $A_{1,0} = 1 + B_{1,0}$, which gives $A_{1,0} = 1 + A_{1,0} - A_{1,1}$ after applying Equation 7.6. Therefore, $A_{1,1} = 1$, and thus

$$B_{1,1} = 1. \tag{7.14}$$

Comparing the $\mathrm{e}^{r_s} r_s^s$ $(s > 1)$ term in both sides of Equation 7.10 gives $A_{s,s} = B_{s,s}$, which is again tautology according to Equation 7.6. Comparing the $\mathrm{e}^{r_s} r_s^l$ ($s > 1$ and $0 \leq l < s$) term gives

$$A_{s,l} = \frac{s}{s-1} B_{s-1,l} + B_{s,l}.$$

By Equation 7.6, we have $A_{s,l} = B_{s,l} + B_{s,l+1}$ if $0 \leq l < s$. Therefore,

$$B_{s,l+1} = \frac{s}{s-1} B_{s-1,l} \qquad (s = 2, 3, \ldots, \quad l = 0, 1, \ldots, s-1)\,. \tag{7.15}$$

Given Equation 7.12, Equation 7.13, Equation 7.14, and Equation 7.15, it is sufficient to determine all $B_{s,l}$ up to one parameter. Choose the parameter to be $b \coloneqq 1 - B_{1,0}$, and then we can prove using mathematical induction that

$$B_{s,l} = \begin{cases} s, & l = s, \\ s\,(1-b)\,, & l = s-1, \\ (-1)^{s+l}\, qb, & l = 0, 1, \ldots, s-2. \end{cases} \tag{7.16}$$

The first few $B_{s,l}$ are listed in Table 7.1.

One can easily get that $b = 0$ by comparing with with results in Equation 6.2, Equation 6.4, or Equation 6.5. If we had not used the ADM to find them before using the method of undetermined coefficients, there is also a way to find b by comparing with Equation 4.8. In this approach, we first expand the left-hand side of Equation 4.8 using Equation 7.1:

$$\int_0^1 g(P, x)\,\mathrm{d}x = \sum_{q=1}^{\infty} \int_{1/(q+1)}^{1/q} g^{(q)}(P, x)\,\mathrm{d}x = \sum_{q=1}^{\infty} \int_{1/(q+1)}^{1/q} \sum_{s=1}^{q} \Delta g_s(P, x)\,\mathrm{d}x.$$

Table 7.1: The first few $B_{s,l}$ given $b \coloneqq 1 - B_{1,0}$. The actual value of b is $b = 0$.

$B_{s,l}$	$l=0$	1	2	3	4	$\cdots$
$s=1$	$1-b$	1				
2	$2b$	$2-2b$	2			
3	$-3b$	$3b$	$3-3b$	3		
4	$4b$	$-4b$	$4b$	$4-4b$	4	
$\vdots$						$\ddots$

Noticing that

$$\sum_{q=1}^{\infty} \int_{1/(q+1)}^{1/q} \sum_{s=1}^{q} = \sum_{s=1}^{\infty} \sum_{q=s}^{\infty} \int_{1/(q+1)}^{1/q} = \sum_{s=1}^{\infty} \int_{0}^{1/s},$$

we have

$$\int_0^1 g(P,x)\,\mathrm{d}x = \sum_{s=1}^{\infty} \int_0^{1/s} \Delta g_s(P,x)\,\mathrm{d}x = \sum_{s=1}^{\infty} \int_0^{1/s} (-1)^s\, \mathrm{e}^{r_s} \ln P \sum_{j=0}^{s} \frac{A_{s,j}}{j!} r_s^j\,\mathrm{d}x,$$

where the last step is from Equation 7.2. Change the integration variable to $w \coloneqq \mathrm{e}^{r_s}$, and then we can use Equation 7.4 and Equation 7.6 to find that

$$\int_0^1 g(P,x)\,\mathrm{d}x = \sum_{s=1}^{\infty} \frac{(-1)^s}{s} \left(B_{s,0} - P^{-1} \sum_{l=0}^{s} \frac{B_{s,l}}{l!} (\ln P)^l \right).$$

Substitute Equation 7.16 into the equation above. Using a method similar to how we calculated the sum resulting to Equation 6.2, we get

$$\int_0^1 g(P,x)\,\mathrm{d}x = P^{-1} - 1 + b \ln P.$$

Comparing with Equation 4.8 gives $b = 0$.

Substituting $b = 0$ into Equation 7.16 gives the values of $B_{s,l}$, and we can then get

the values of $A_{s,j}$ by relating them with Equation 7.6. The result is

$$A_{s,j} = \begin{cases} s, & j = s, s-2, \\ 2s, & j = s-1, \\ 0, & j = 0, 1, \ldots, s-3. \end{cases} \tag{7.17}$$

It can be proven by mathematical induction. The first few values of $A_{s,j}$ are listed in Table 7.2.

Table 7.2: The first few $A_{s,j}$.

$A_{s,j}$	$j=0$	1	2	3	4	$\cdots$
$s=1$	2	1				
2	2	4	2			
3	0	3	6	3		
4	0	0	4	8	4	
$\vdots$						$\ddots$

Substituting Equation 7.17 into Equation 7.2, and we have

$$\Delta g_s(P,x) = \begin{cases} -\mathrm{e}^{r_1} \ln P \left(2 + r_1\right), & s = 1, \\ \dfrac{s\,(-1)^s}{(s-2)!} \mathrm{e}^{r_s} r_s^{s-2} \ln P \left(1 + \dfrac{2}{s-1} r_s + \dfrac{1}{s\,(s-1)} r_s^2\right), & s = 2, 3, \ldots \end{cases} \tag{7.18}$$

Substituting Equation 7.18 into Equation 7.1, and we have

$$\frac{g^{(q)}(P,x)}{-\ln P} = \mathrm{e}^{r_1}\left(2 + r_1\right) - \sum_{s=2}^{q} \frac{s\,(-1)^s}{(s-2)!} \mathrm{e}^{r_s} r_s^{s-2} \left(1 + \frac{2}{s-1} r_s + \frac{1}{s\,(s-1)} r_s^2\right). \tag{7.19}$$

Notice that although we assumed $q > 1$ in this section, Equation 7.19 is also applicable to $q = 1$.

We are now only one step from the full solution of $f(P,x)$. Glue together the pieces of different orders to get the full solution of $g(P,x)$, and then one can get $f(P,x)$ using

Equation 4.6. The result is

$$f(P=0,x)=\delta(x)\,,$$
$$f(0<P\leq 1,x)=\begin{cases}\ln P\sum_{s=2}^{\lceil 1/x\rceil-1}\frac{s\,(-1)^s}{(s-2)!}P^{sx}r_s^{s-2}\left(1+\frac{2}{s-1}r_s+\frac{1}{s(s-1)}r_s^2\right) & 0<x\leq 1,\\ \quad +P\delta(x-1)-P^x\ln P\,(2+r_1)\,, & \\ 0, & \text{otherwise,}\end{cases} \tag{7.20}$$

where $r_s := \ln P^{sx-1}$. The reason for using the ceiling function instead of the floor function is in Chapter 8. A plot of this function is shown in Figure 7.2, and a compare of this function with the discrete case is shown in Figure 7.3. We can see that for $n = 50$, it is already close to the continuum limit.

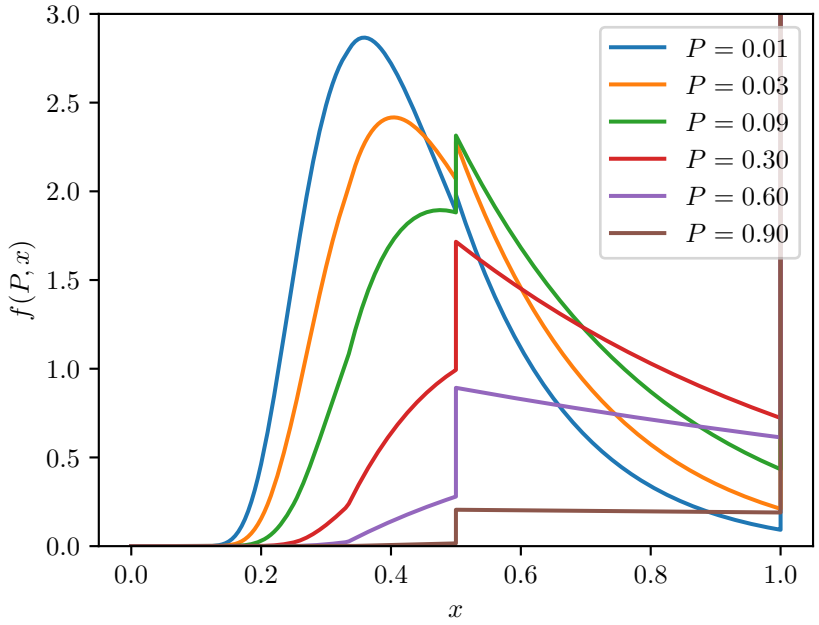


Figure 7.2: Plot of $f(P,x)$ for various values of P.

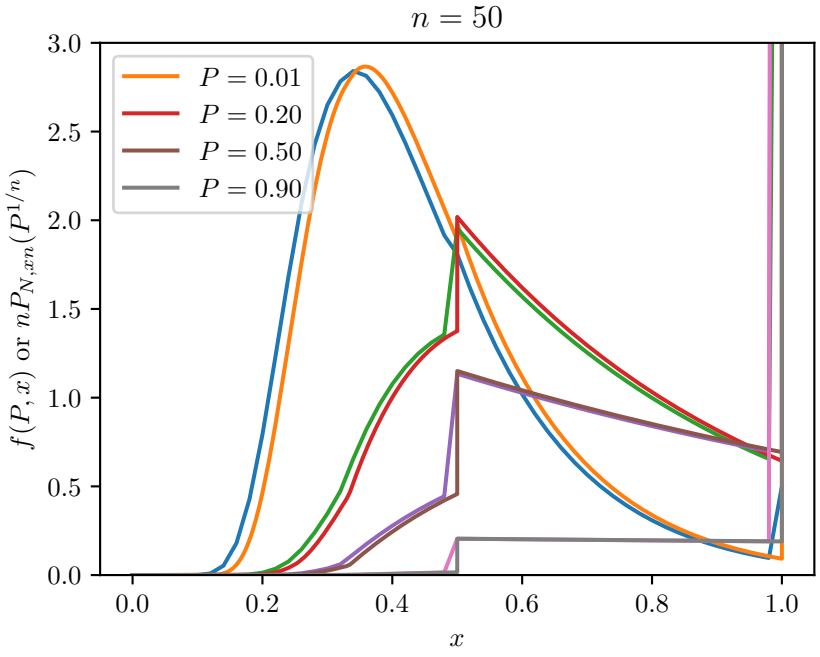


Figure 7.3: Comparing $f(P, x)$ and $P_{n,xn}(p)$ in the same plot for $n = 50$. The lines for $P_{n,xn}(p)$ are not legended.

Chapter 8

Singularities in the distribution function

From Equation 7.19, we see that $g^{(q)}(P,x)$ is a smooth function. However, $g(P,x)$ is not smooth everywhere: it is not infinitely smooth at $x = 1/q$ ($q > 1$) because it is $g^{(q)}(P,x)$ to the left of $x = 1/q$ while it is $g^{(q-1)}(P,x)$ to the right of $x = 1/q$. From Figure 7.2 and as we foreshadowed in the end of Chapter 3, $x = 1/q$ is a singularity, and it is a discontinuity in the $(q-2)$th derivative w.r.t. x of $f(P,x)$. In this section, we will show that the jth derivative w.r.t. x of $f(P,x)$ is continuous at $x = 1/q$ if $j < q-2$ but is discontinuous if $j = q-2$.

The difference between the left-limit and the right-limit of the jth derivative w.r.t. x of $f(P,x)$ at $x = 1/q$ is

$$D_{q,j} \coloneqq \lim_{\varepsilon \to 0^+} \left. \frac{\partial^j f(P,x)}{\partial x^j} \right|_{x=1/q-\varepsilon}^{1/q+\varepsilon} = P \left. \frac{\partial^j g^{(q-1)}(P,x)}{\partial x^j} \right|_{x=1/q} - P \left. \frac{\partial^j g^{(q)}(P,x)}{\partial x^j} \right|_{x=1/q}.$$

See Equation 4.6 for the factor of P. From Equation 7.1, we have

$$D_{q,j} = -P \left. \frac{\partial^j \Delta g_q(P,x)}{\partial x^j} \right|_{x=1/q}.$$

As complicated as $\Delta g_q(P,x)$ is, taking its derivative w.r.t. x is potentially difficult. However, from Equation 7.18, we know that it is simpler to be expressed as a function of P and r_q. Noticing that $r_q \coloneqq \ln P^{qx-1}$ is linear in x, we have

$$\frac{\partial}{\partial x} = q \ln P \frac{\partial}{\partial r_q}, \tag{8.1}$$

where $\partial/\partial r_q$ acting on a function of P and x is understood as taking the derivative w.r.t. r_q while keeping P constant. We then have

$$D_{q,j} = -\left(q \ln P\right)^j P \left.\frac{\partial^j \Delta g_q(P,x)}{\partial r_q^j}\right|_{r_q=0}.$$

It is then straightforward to see that this expression is zero when $j < q-2$ and nonzero when $j = q-2$ as we look at Equation 7.19: if we expand $\Delta g_q(P,x)$ as a Talor series in r_q around $r_q = 0$, then the lowest-order nonzero term in the expansion is the r_q^{q-2} term. Reading out the coefficient of that term gives (a factor of) $D_{q,q-2}$. Therefore,

$$\begin{aligned} D_{q,j} &= 0 \qquad (j = 0, 1, \ldots, q-3)\,, \\ D_q \coloneqq D_{q,q-2} &= (-q \ln P)^{q-1} P. \end{aligned}$$

Other than the singularities at $x = 1/q$ and $x = 1$, there is another more essential singularity at $x = 0$. The behavior of $f(P,x)$ is pathological at that point: $f(P,x)$ is infinitely smooth at $x = 0$, but there are infinitely many singularities (those at $x = 1/q$) where $f(P,x)$ is not infinitely smooth in any neighborhood of $x = 0$. To show the smoothness, define

$$g^{(\infty)}(P,x) \coloneqq \sum_{s=1}^{\infty} \Delta g_s(P,x)\,.$$

From Equation 7.18, we know that

$$\lim_{s\to\infty} \frac{\Delta g_{s+1}(P,x)}{\Delta g_s(P,x)} = \lim_{s\to\infty} \frac{\mathrm{e}^{r_{s+1}} \left(-r_{s+1}\right)^{s+1}}{s\mathrm{e}^{r_s} \left(-r_s\right)^s} = -\mathrm{e}xP^x \ln P \leq 1,$$

where the equality holds if and only if $x = -1/\ln P$. Therefore, $g^{(\infty)}(P,x)$ converges absolutely if $x \neq -1/\ln P$ (the convergence at $x = -1/\ln P$ cannot be known through this

argument, but it is not important; in fact, it diverges: using Stirling's approximation, one can show that $\Delta g_s(P, -1/\ln P) \sim -\ln P/\sqrt{2\pi s})$. It converges uniformly in a neighborhood of $x = 0$. Then, because each summand in $g^{(\infty)}(P, x)$ is analytic in a neighborhood of $x = 0$, $g^{(\infty)}(P, x)$ is also analytic (and thus infinitely smooth) in a neighborhood of $x = 0$. Therefore, the infinite series

$$\partial_x^j g^{(\infty)}(P, x) = \sum_{s=1}^{\infty} \partial_x^j \, \Delta g_s(P, x)$$

also converges uniformly and absolutely and is analytic in a neighborhood of $x = 0$. Its tail, starting at index s_0,

$$\partial_x^j R_{s_0}(P, x) \coloneqq \sum_{s=s_0}^{\infty} \partial_x^j \, \Delta g_s(P, x)$$

then converges uniformly to zero as $s_0 \to \infty$. Now, notice that

$$\partial_x^j g^{(\infty)}(P, x) - \partial_x^j g(P, x) = \partial_x^j R_{\lceil 1/x \rceil}(P, x)$$

(for the derivative to be guaranteed to exist, x needs to be in a small enough neighborhood of $x = 0$ so that $x < 1/(j + 2)$) has its starting index $\lceil 1/x \rceil \to \infty$ as $x \to 0^+$, so

$$\lim_{x \to 0^+} \partial_x^j g(P, x) = \partial_x^j g^{(\infty)}(P, 0)$$

(where $\partial_x^j \cdot(0)$ is an abbreviation of $\partial_x^j \cdot(x)|_{x=0}$) exists. Then, we will prove using mathematical induction that $\partial_x^j g(P, 0)$ exists and is equal to this limit. The base case $j = 0$ is true because $g^{(\infty)}(P, 0) = 0 = g(P, 0)$, where the first equal sign is derived by direct evaluation of the series. For the inductive step, we have

$$\partial_x^{j+1} g(P, 0) = \lim_{x \to 0^+} \frac{\partial_x^j g(P, x) - \partial_x^j g(P, 0)}{x} = \lim_{x \to 0^+} \frac{\partial_x^j g(P, x) - \partial_x^j g^{(\infty)}(P, 0)}{x}.$$

Using L'Hôpital's rule, we then have

$$\partial_x^{j+1} g(P, 0) = \lim_{x \to 0^+} \partial_x^{j+1} g(P, x)\,,$$

and this finishes the proof. The smoothness of $g(P,x)$ at $x=0$ then implies the smoothness of $f(P,x)$ at that point.

In fact, all orders of derivatives of $f(P,x)$ at $x=0$ vanish. To prove this, first rewrite Equation 7.18 as

$$\Delta g_s(P,x) = \frac{s\,(-1)^s}{s!}\ln P\,\frac{\partial^2}{\partial r_s^2}\left(\mathrm{e}^{r_s} r_s^s\right). \tag{8.2}$$

Act ∂_x^j on both sides of it and evaluate at $x=0$. Using Equation 8.1, we have

$$\partial_x^j\,\Delta g_s(P,0) = \frac{s^{j+1}\,(-1)^s}{s!}\,(\ln P)^{j+1}\,\frac{\partial^{j+2}}{\partial r_s^{j+2}}\left(e^{r_s} r_s^s\right)\Big|_{r_s=-\ln P}.$$

Notice that now r_s is just a bound variable, so we can rename it to z. Sum over s, and we have

$$\partial_x^j g^{(\infty)}(P,0) = (\ln P)^{j+1}\,\frac{\partial^{j+2}}{\partial z^{j+2}}\left(\mathrm{e}^z\sum_{s=1}^{\infty}\frac{s^{j+1}\,(-z)^s}{s!}\right)\Big|_{z=-\ln P}.$$

Notice that this is exactly the Touchard polynomial [12]

$$T_{j+1}(-z) = \mathrm{e}^z\sum_{s=0}^{\infty}\frac{s^{j+1}\,(-z)^s}{s!},$$

which is a polynomial of degree $j+1$. Therefore, its $(j+2)$th derivative vanishes, and we then have $\partial_x^j g^{(\infty)}(P,0)=0$.

After the discussions about the singularities, we see that the only points where $f(P,x)$ is discontinuous are $x=1/2$ and $x=1$. Whether $f(P,1/2)$ is taken to be the left limit or the right limit determines whether the upper limit of the summed index is $\lceil 1/x\rceil - 1$ or $\lfloor 1/x\rfloor$ in Equation 7.20. The two different choices result in the same probability distribution because probability distributions are only defined up to a zero difference almost everywhere. However, it matters as in the definition in Equation 4.1. Looking at Equation 4.7, one may see that the discontinuity at $x=1/2$ is due to the δ function in the integrand. Therefore, whether g at $x=1/2$ is $g^{(1)}$ or $g^{(2)}$ depends on whether the δ function is within the integrated interval. If it is, then g at $x=1/2$ is $g^{(1)}$; otherwise, it is $g^{(2)}$. The inclusion of the δ function in the integrated interval corresponds to the

inclusion of the highest term in the summation in Equation 3.1. Because both k and $k-1$ are at least as large as $n-k-1$ when $n=2k$, the highest term in the summation can be reached, so the δ function is within the integrated interval. Therefore, g at $x=1/2$ is $g^{(1)}$. This means $g(P,x)=g^{(\lceil 1/x\rceil -1)}(P,x)$ and hence the ceiling function in Equation 7.20.

Chapter 9

Integrated cumulative distribution function

The integrated cumulative distribution function is defined as the integral of the cumulative distribution function or the twice integral of the probability density function:

$$F(P,x) \coloneqq \int_0^x \mathrm{d}x' \int_0^{x'} \mathrm{d}x''\, f(P,x'')\,. \tag{9.1}$$

The integrated cumulative distribution function has its application in decision theory in that it can be used to measure the second-order stochastic dominance, which is in turn used to measure risks [16]. Our interest in the integrated cumulative distribution function here is not because of its applications, but because of the potential conciseness in expressing the probability distribution embodied by the simpler form of Equation 8.2 than Equation 7.18.

Using Equation 8.1, we can rewrite Equation 8.2 as

$$\Delta g_s(P,x) = \partial_x^2\, \Delta G_s(P,x)\,,$$

where

$$\Delta G_s(P,x) \coloneqq \frac{(-1)^s}{s!\, s \ln P} \mathrm{e}^{r_s} r_s^s. \tag{9.2}$$

From Equation 9.1, Equation 4.6, and Equation 7.1, we then have

$$F(P, 0 < x < 1) = P \sum_{s=1}^{\lceil 1/x \rceil - 1} \Delta G_s(P, x) - \mu - \kappa x, \tag{9.3}$$

where μ and κ are defined so that $F(P, x)$ satisfies the boundary conditions

$$F(P, 0) = 0, \qquad \partial_x F(P, 0) = 0$$

imposed by Equation 9.1. We are free to choose between $\lfloor 1/x \rfloor$ and $\lceil 1/x \rceil - 1$ for the upper limit in s because $\Delta G_s(P, 1/s) = 0$. Now, let us consider $F(P, x > 1)$. To account for the δ function peak in $f(P, x)$ at $x = 1$ (see Equation 4.6), we need a jump in $\partial_x F(P, x)$ at $x = 1$ as large as P. First calculate the left limit ($x \to 1^-$). From Equation 9.3, an easy calculation gives

$$\partial_x F\big(P, 1^-\big) = P\, \partial_x\, \Delta G_1(P, 1) - \kappa = -P - \kappa.$$

To have a jump as large as P, we need

$$\partial_x F\big(P, 1^+\big) = -\kappa. \tag{9.4}$$

This is exactly what Equation 9.3 would yield if we simply use the same expression for $x > 1$. Therefore,

$$F(P, x) = P \sum_{s=1}^{\lfloor 1/x \rfloor} \Delta G_s(P, x) - \mu - \kappa x. \tag{9.5}$$

Calculation of κ is a straghtforward evaluation:

$$\kappa \coloneqq P \sum_{s=1}^{\infty} \partial_x\, \Delta G_s(P, 0) = P \sum_{s=1}^{\infty} \frac{(-1)^s}{s!\, s \ln P} \frac{s\,(\ln P - s)\,(-\ln P)^s}{P} = -1. \tag{9.6}$$

Calculation of μ involves evaluating this series:

$$\mu \coloneqq P \sum_{s=1}^{\infty} \Delta G_s(P, 0) = P \sum_{s=1}^{\infty} \frac{(-1)^s}{s!\, s \ln P} \frac{(-\ln P)^s}{P}. \tag{9.7}$$

This series is closely related to the exponential integral [17, p. 229]

$$\operatorname{Ei} z \coloneqq \int_{-\infty}^{z} \frac{\mathrm{e}^t}{t}\,\mathrm{d}t = \gamma + \ln|z| + \sum_{s=1}^{\infty} \frac{z^s}{s!\,s}, \tag{9.8}$$

where γ is the Euler–Mascheroni constant, and z is nonzero and real. We therefore has $\mu = \left(\operatorname{Ei} \ln P - \gamma - \ln(-\ln P)\right)/\ln P$. We can simplify it using the logarithmic integral [17, p. 228]

$$\operatorname{li} z \coloneqq \int_0^z \frac{\mathrm{d}t}{\ln t} = \operatorname{Ei} \ln z,$$

so

$$\mu = \frac{\operatorname{li} P - \gamma - \ln(-\ln P)}{\ln P}. \tag{9.9}$$

The quantity μ has statistical meaning. Consider the mean $\mu_1(P)$ of the distribution:

$$\mu_1(P) \coloneqq \int_0^{\infty} x f(P,x)\,\mathrm{d}x = \int_0^{1^+} x\,\partial_x^2 F(P,x)\,\mathrm{d}x.$$

Integrate by parts, and then we have

$$\mu_1(P) = x\,\partial_x F(P,x)\big|_{x=0}^{1^+} - \int_0^{1^+} \partial_x F(P,x)\,\mathrm{d}x = \left(x\,\partial_x F(P,x) - F(P,x)\right)\big|_{x=0}^{1^+}.$$

From Equation 9.5 and Equation 9.4, we have

$$\mu_1(P) = -\kappa - (-\mu - \kappa) = \mu.$$

Therefore, μ is exactly the mean of the distribution. A plot of how the mean changes with P is shown in Figure 9.1.

Substitute Equation 9.6, Equation 9.9, and Equation 9.2 into Equation 9.5, and we have

$$F(P,x) = \frac{P}{\ln P} \sum_{s=1}^{\lfloor 1/x \rfloor} \frac{(-1)^s}{s!\,s} \mathrm{e}^{r_s} r_s^s - \frac{\operatorname{li} P - \gamma - \ln(-\ln P)}{\ln P} + x. \tag{9.10}$$

This function is plotted for various values of P in Figure 9.2.

Similarly to what was shown in Chapter 8, $F(P,x)$ also has singularities. Namely, for each of $q = 1, 2, \ldots$, there is a singularity at $x = 1/q$, at which point $\partial_x^q F(P,x)$ jumps

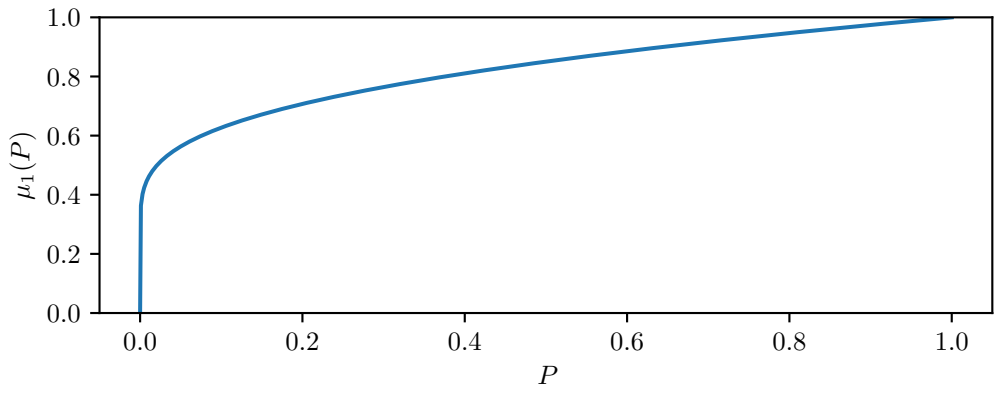


Figure 9.1: The dependence on P of the mean $\mu_1(P)$ of the distribution with probability density $f(P, x)$.

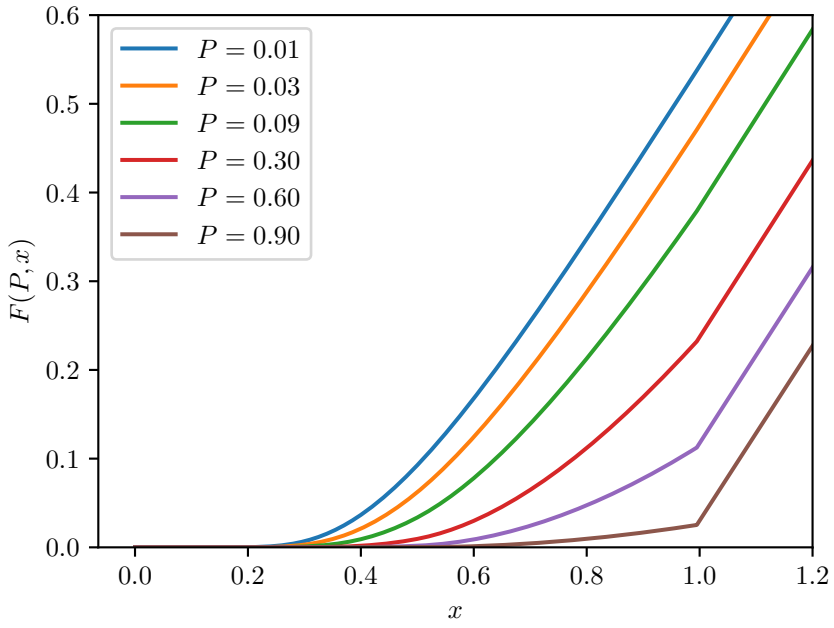


Figure 9.2: Plot of $F(P, x)$ for various values of P.

by D_q but all lower derivatives w.r.t. x are continuous. At $x = 0$, $F(P, x)$ is infinitely smooth, and all orders of its derivatives w.r.t. x vanish.

Chapter 10

Broken stick distribution

The broken stick problem states as follows. Given a stick of unit length, if it is broken at random k points (independently and uniformly distributed along the stick) $X_1, \ldots, X_k$, it will be divided into $k+1$ segments, whose lengths are

$$S_0 = X_1, \quad S_1 = X_2 - X_1, \quad \ldots, \quad S_k = 1 - X_k.$$

If we sort the lengths in increasing order as $S_{(0)} \leq \cdots \leq S_{(k)}$, then what is the exact distribution of $S_{(k-j)}$, i.e. the length of the jth (zero-based) longest segment?

For reasons of comparing with the results with previous sections, we will mostly focus on the longest segment only. However, in this section, a general result is given. One may find the same method being used elsewhere in literature [8].

First, observe that the broken stick model has the symmetry of exchanging labels on the segments: the joint distribution of $S_0, \ldots, S_k$ is exactly the same as that of $S_{\pi(0)}, \ldots, S_{\pi(k)}$, where π is a permutation of the indices $0, \ldots, k$. This fact will be useful in the derivation of the distribution of $S_{(k-j)}$.

The goal is to find the cumulative distribution function $\mathrm{P}\big[S_{(k-j)} \leq x\big]$ for any $x \geq 0$.

Define binary random variables $I_0, \ldots, I_k$ such that

$$I_l \coloneqq \begin{cases} 1, & S_l > x, \\ 0, & S_l \leq x \end{cases} \qquad (l = 0, \ldots, k).$$

The condition $S_{(k-j)} \leq x$ is equivalent to the condition that there are at most j segments longer than x, so

$$\mathrm{P}\left[S_{(k-j)} \leq x\right] = \mathrm{P}\left[\sum_{l=0}^{k} I_l \leq j\right] = \sum_{m=0}^{j} \mathrm{P}\left[\sum_{l=0}^{k} I_l = m\right]. \tag{10.1}$$

The probability $\mathrm{P}\left[\sum_{l=0}^{k} I_l = m\right]$ equals summing the probability for each way of choosing the m segments among the total $k+1$ segments. The random variables $I_0, \ldots, I_k$ also has the symmetry of exchanging labels, so each term in the summation is equal. Therefore,

$$\mathrm{P}\left[\sum_{l=0}^{k} I_l = m\right] = \binom{k+1}{m} \mathrm{E}\left[\prod_{l=0}^{m-1} I_l \prod_{l=m}^{k} (1 - I_l)\right].$$

Expand the multiplication in the square bracket and use the symmetry again, and we have

$$\mathrm{P}\left[\sum_{l=0}^{k} I_l = m\right] = \binom{k+1}{m} \sum_{i=0}^{k-m+1} \binom{k-m+1}{i} (-1)^i \mathrm{E}\left[\prod_{l=0}^{m+i-1} I_l\right]. \tag{10.2}$$

Now, we need to calculate

$$\mathrm{E}\left[\prod_{l=0}^{s-1} I_l\right] = \mathrm{P}\left[\bigwedge_{l=0}^{s-1} S_l > x\right]. \tag{10.3}$$

We can prove using mathematical induction that

$$\mathrm{P}\left[\bigwedge_{l=0}^{s-1} S_l > x\right] = (1 - sx)^{*k}, \tag{10.4}$$

where the notation z^{*k} means the rectified power function [18, p. 7]:

$$z^{*k} \coloneqq \begin{cases} z^k, & z > 0, \\ 0, & z \leq 0. \end{cases}$$

For the base case,

$$\mathrm{P}[S_0 > x] = \mathrm{P}\left[\bigwedge_{l=1}^{k} X_l > x\right] = (1-x)^{*k}.$$

For the inductive step,

$$\mathrm{P}\left[\bigwedge_{l=0}^{s} S_l > x\right] = \mathrm{P}\left[\bigwedge_{l=1}^{s} S_l > x \,\middle|\, S_0 > x\right] \mathrm{P}[S_0 > x].$$

The second factor is $(1-x)^{*k}$, the same as the base case. For the first factor, since $S_0 > x$ is equivalent to $\bigwedge_{l=1}^{k} X_l > x$, the situation $\bigwedge_{l=1}^{s} S_l > x$ is equivalent to the situation in the inductive assumption but with the points distributed on an interval of length $1-x$ instead of unit length. Therefore,

$$\mathrm{P}\left[\bigwedge_{l=0}^{s} S_l > x\right] = \begin{cases} \left(1 - s\dfrac{x}{1-x}\right)^{*k} (1-x)^{*k}, & 1-x > 0, \\ 0, & 1-x \le 0 \end{cases} = (1-(s+1)\,x)^{*k}.$$

This completes the proof.

Substitute Equation 10.4 into Equation 10.3, and then substitute into Equation 10.2, and finally sbustitute into Equation 10.1, and we have the broken stick distribution

$$\mathrm{P}\big[S_{(k-j)} \le x\big] = \sum_{m=0}^{j} \binom{k+1}{m} \sum_{i=0}^{k-m+1} \binom{k-m+1}{i} (-1)^i \,(1-(m+i)\,x)^{*k}.$$

Specially, the distribution of the length of the longest segment is

$$\mathrm{P}\big[S_{(k)} \le x\big] = \sum_{s=0}^{k+1} \binom{k+1}{s} (-1)^s \,(1-sx)^{*k}. \tag{10.5}$$

One is encouraged to check that Equation 10.5 is still valid when $x \ge 1$ and when $k = 0$.

We can integrate or take the derivative w.r.t. x with Equation 10.5 to get the probability density function and the integrated cumulative distribution function. Notice that we have (for derivatives at points where the function is not differentiable, the one-sided limit

is taken as the value there, the reason of which will be clear in the end of Section 11.1)

$$\int_0^x (1-sx')^{*k}\,\mathrm{d}x' = \begin{cases} \dfrac{1-(1-sx)^{*k+1}}{s\,(k+1)}, & s>0, \\ x, & s=0, \end{cases}$$

$$\partial_x\,(1-sx)^{*k} = \begin{cases} -sk\,(1-sx)^{*k-1}, & k>0, \\ -\delta\left(x-\dfrac{1}{s}\right), & k=0\wedge s>0, \\ 0, & k=0\wedge s=0. \end{cases}$$

Therefore, the probability density function is

$$f_k(x) \coloneqq \partial_x\,\mathrm{P}\left[S_{(k)}\le x\right] = \begin{cases} -\displaystyle\sum_{s=1}^{k+1} \frac{(k+1)!\,k}{(s-1)!\,(k-s+1)!}\,(-1)^s\,(1-sx)^{*k-1}, & k>0, \\ \delta(x-1), & k=0, \end{cases} \tag{10.6}$$

and the integrated cumulative distribution function is

$$F_k(x) \coloneqq \int_0^x \mathrm{P}\left[S_{(k)}\le x'\right]\mathrm{d}x' = \sum_{s=1}^{k+1} \frac{k!\,(-1)^s}{s!\,s\,(k-s+1)!}\left(1-(1-sx)^{*k+1}\right)+x. \tag{10.7}$$

The two functions are plotted in Figure 10.1 for various values of k.

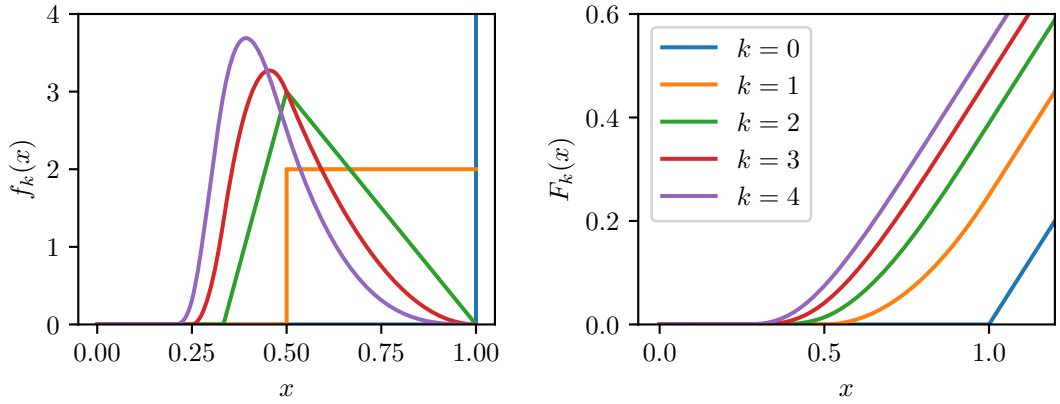


Figure 10.1: Plot of $f_k(x)$ and $F_k(x)$ for various values of k. The legend is shared.

Chapter 11

Generating function of the broken stick distribution

There is a relation between the broken stick model and the continuum limit of the longest run statistics. For starters, in the continuum limit concerned in Chapter 4, the occurrences of F's form a Poisson point process on the unit interval. The relation between the Bernoulli process and the Poisson process is textbook-level and is also the reason why the Poisson process is named after Poisson [19]. Here, we give an explicit derivation of this relation.

According to the binomial distribution, the probability of having k failures out of n trails is given by

$$\binom{n}{k} p^{n-k} (1-p)^k = \binom{n}{k} P^{1-k/n} \left(1 - P^{1/n}\right)^k.$$

The binomial coefficient is a polynomial in n, whose leading term is $n^k/k!$. Therefore, the limit in $n \to \infty$ is given by

$$\lim_{n\to\infty} \binom{n}{k} p^{n-k} (1-p)^k = \frac{P}{k!} \left(\lim_{n\to\infty} nP^{-1/n} \left(1 - P^{1/n}\right)\right)^k = \frac{P\,(-\ln P)^k}{k!},$$

which is exactly the Poisson distribution with parameter $\lambda \coloneqq -\ln P$.

To relate the distributions $f(P,x)$ and $f_k(x)$, simply average over $f_k(x)$ with different values of k weighted by the Poisson distribution:

$$f\left(\mathrm{e}^{-\lambda},x\right)=\mathrm{e}^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!}f_k(x)\,,\qquad f(P,x)=P\sum_{k=0}^{\infty}\frac{(-\ln P)^k}{k!}f_k(x)\,. \tag{11.1}$$

This means that $f\left(\mathrm{e}^{-\lambda},x\right)\mathrm{e}^{\lambda}$ as a function of λ (or $f(P,x)/P$ as a function of $-\ln P$) is the exponential generating function [20, p. 95] of $f_k(x)$ as a sequence indexed by k. Similarly, the integrated cumulative distribution functions also have this relation, derived simply by integrating both sides of Equation 11.1 twice:

$$F\left(\mathrm{e}^{-\lambda},x\right)=\mathrm{e}^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!}F_k(x)\,,\qquad F(P,x)=P\sum_{k=0}^{\infty}\frac{(-\ln P)^k}{k!}F_k(x)\,. \tag{11.2}$$

Because a sequence and its exponential generating function can be converted to and from each other (summation converts the sequence to the function and Taylor series expansion converts the function to the sequence), this means that the two models, the continuum limit of the longest run statistics and the longest segment of the broken stick, can derive each other.

In Section 11.1, we derive $f(P,x)$ from $f_k(x)$, and in Section 11.2, we derive $f_k(x)$ from $f(P,x)$.

11.1 Finding the generating function

Because the integrated cumulative distribution functions have simpler forms, we will work with Equation 11.2 first although Equation 11.1 works just as fine. In this section, we will recover Equation 9.10 from Equation 10.7.

Substitute Equation 10.7 into Equation 11.2, and then we have

$$\frac{F(P,x)}{P}=\sum_{k=0}^{\infty}\frac{\lambda^k}{k!}\left(\sum_{s=1}^{k+1}\frac{k!\,(-1)^s}{s!\,s\,(k-s+1)!}\left(-\left(1-sx\right)^{*k+1}+1\right)+x\right).$$

Expand to get all three terms and exchange the summation order of s and k, and it becomes

$$\frac{F(P,x)}{P} = \sum_{s=1}^{\infty} -\frac{(-1)^s}{s!\,s} \sum_{k=s-1}^{\infty} \frac{\lambda^k}{(k-s+1)!} (1-sx)^{*k+1} + \sum_{s=1}^{\infty} \frac{(-1)^s}{s!\,s} \sum_{k=s-1}^{\infty} \frac{\lambda^k}{(k-s+1)!} + x \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}.$$

The third term is standard. For the other two terms, change the summation variable $k \to k-s+1$, and we have

$$\frac{F(P,x)}{P} = \sum_{s=1}^{\infty} -\frac{(-1)^s}{s!\,s} \sum_{k=0}^{\infty} \frac{\lambda^{k+s-1}}{k!} (1-sx)^{*k+s} + \sum_{s=1}^{\infty} \frac{(-1)^s}{s!\,s} \sum_{k=0}^{\infty} \frac{\lambda^{k+s-1}}{k!} + x\mathrm{e}^{\lambda}.$$

Rearrange the factors and calculate out the sums over k. Note that $r_s = (1-sx)\,\lambda$, so we have

$$\frac{F(P,x)}{P} = \sum_{s=1}^{\infty} -\frac{(-1)^s\, r_s^{*s} \lambda^{-1}}{s!\,s} \mathrm{e}^{r_s} + \mathrm{e}^{\lambda} \sum_{s=1}^{\infty} \frac{(-1)^s}{s!\,s} \lambda^{s-1} + x\mathrm{e}^{\lambda}.$$

The summand in the first term is exactly $\Delta G_s(P,x)$ (Equation 9.2) when $x < 1/s$, and the sum in the second term is exactly μ (Equation 9.7). Therefore, Equation 9.5 is recovered, which is desired.

One can also recover Equation 7.20 directly using the same approach. First substitute Equation 10.6 into Equation 11.1, and then exchange the order of summation over k and s. The summation over k can be carried out, and then the desired formula can be recovered. The only subtlety that is not involved in the discussion with $F_k(x)$ is whether we can recover the correct value of $f(P,1/2)$. From the discussion in the end of Chapter 8, $f(P,x)$ is right-continuous but not left-continuous at $x = 1/2$. The only k that makes $f_k(x)$ have a discontinuity at $x = 1/2$ is $k = 1$ due to the term $(1-2x)^{*0}$. It is right-continuous by the definition of the rectified power function, so it is consistent with the behavior of $f(P,x)$ at $x = 1/2$.

11.2 Expanding the generating function

In this section, we will recover Equation 10.7 from Equation 9.10. Looking at the series expansion in Equation 11.2, we have

$$F_k(x) = \partial_\lambda^k \frac{F(P,x)}{P}\bigg|_{\lambda=0},$$

where ∂_λ^k means taking the kth derivative w.r.t. λ keeping x constant. From Equation 9.5, we have

$$F_k(x) = \sum_{s=1}^{\lfloor 1/x \rfloor} \partial_\lambda^k \, \Delta G_s(P,x)\big|_{\lambda=0} - \partial_\lambda^k \frac{\mu}{P}\bigg|_{\lambda=0} + x \, \partial_\lambda^k \mathrm{e}^\lambda\big|_{\lambda=0}. \tag{11.3}$$

Because $r_s = (1-sx)\,\lambda$, we have

$$\partial_\lambda = (1-sx)\,\frac{\partial}{\partial r_s},$$

where $\partial/\partial r_s$ means taking derivative w.r.t. r_s while keeping x constant. Therefore, acting ∂_λ^k on Equation 9.2 gives

$$\partial_\lambda^k \, \Delta G_s(P,x) = -\frac{(-1)^s\,(1-sx)^{k+1}}{s!\,s}\frac{\partial^k}{\partial r_s^k}\left(\mathrm{e}^{r_s} r_s^{s-1}\right).$$

Take its value at $\lambda = 0$ (i.e. $r_s = 0$), and we have

$$\partial_\lambda^k \, \Delta G_s(P,x)\big|_{\lambda=0} = -\frac{(-1)^s\,(1-sx)^{k+1}}{s!\,s}\,\partial_z^k \left(\mathrm{e}^z z^{s-1}\right)\big|_{z=0}.$$

We simply need to find the kth order coefficient in the Taylor series expansion of $\mathrm{e}^z z^{s-1}$ at $z=0$. Obviously, it is 0 if $k < s-1$, and it is $1/(k-s+1)!$ if $k \geq s-1$. In other words,

$$\partial_z^k \left(\mathrm{e}^z z^{s-1}\right)\big|_{z=0} = \begin{cases} 0, & k < s-1, \\ \dfrac{k!}{(k-s+1)!}, & k \geq s-1. \end{cases}$$

Therefore,

$$\partial_\lambda^k \, \Delta G_s(P,x)\big|_{\lambda=0} = \begin{cases} 0, & k < s-1, \\ -\dfrac{k!\,(-1)^s\,(1-sx)^{k+1}}{s!\,s\,(k-s+1)!}, & k \geq s-1. \end{cases} \tag{11.4}$$

For the second term in Equation 11.3, we can act ∂_λ^k on Equation 9.7 to get

$$\partial_\lambda^k \frac{\mu}{P}\Big|_{\lambda=0} = \sum_{s=1}^{\infty} \partial_\lambda^k \, \Delta G_s(P,0)\big|_{\lambda=0} \, .$$

The summand is then the special case of Equation 11.4 for $x = 0$. Therefore,

$$\partial_\lambda^k \frac{\mu}{P}\Big|_{\lambda=0} = \sum_{s=1}^{k+1} -\frac{k!\,(-1)^s}{s!\,s\,(k-s+1)!}. \tag{11.5}$$

The third term in Equation 11.3 is easily just x. Therefore, if we substitute Equation 11.4 and Equation 11.5 into Equation 11.3, we get

$$F_k(x) = -\sum_{s=1}^{\min(\lfloor 1/x \rfloor, k+1)} \frac{k!\,(-1)^s\,(1-sx)^{k+1}}{s!\,s\,(k-s+1)!} + \sum_{s=1}^{k+1} \frac{k!\,(-1)^s}{s!\,s\,(k-s+1)!} + x.$$

This recovers Equation 10.7.

Chapter 12

Gumbel distribution as a limiting case

The Gumbel distribution is a continuous distribution used commonly in statistics. The maximum value in a sample of random variables following an exponential distribution minus the natural logarithm of the sample size approaches the Gumbel distribution as the sample size tends to infinity. The probability density function of a random variable following the standard Gumbel distribution is by definition $\exp(-y - \mathrm{e}^{-y})$ (as a function of y) [21]. Integrating it twice, we can get the integrated cumulative distribution function of the standard Gumbel distribution:

$$G(y) \coloneqq \int_{-\infty}^{y} \mathrm{d}y' \int_{-\infty}^{y'} \mathrm{d}y'' \, \exp\bigl(-y - \mathrm{e}^{-y}\bigr) = -\operatorname{Ei}\bigl(-\mathrm{e}^{-y}\bigr)\,,$$

where $\operatorname{Ei} z$ is the exponential integral function, whose definition is given in Equation 9.8. More general Gumble distribution is parametrized by two parameters $\bar{\mu}$ and β, and $(X - \bar{\mu})/\beta$ follows the standard Gumbel distribution if X follows the Gumbel distribution with parameters $\bar{\mu}$ and β. The mean of this distribution is $\bar{\mu} + \beta\gamma$, where γ is the Euler–Mascheroni constant.

In this section, we will show that the distribution of the longest run statistics in the continuum limit tends to the Gumbel distribution in a certain limit. To elaborate, we have

$$\lim_{\lambda\to\infty} \lambda F\left(\mathrm{e}^{-\lambda}, \frac{y+\ln\lambda}{\lambda}\right) = G(y)\,. \tag{12.1}$$

Keeping using $\lambda \coloneqq -\ln P$, we know that the limit $\lambda \to \infty$ is equivalent to $P \to 0$. Figure 12.1 plots both sides of Equation 12.1 for various values of λ. Intuitively, one can see from Figure 7.2 that the probability density function $f(P, x)$ looks smoother as P decreases. The reason of this tendency is exactly that the Gumbel distribution is the smooth limit of the distribution as P tends to zero. To elaborate, $f(P, x)$ is approximately the probability density function of the Gumbel distribution with parameters $\bar{\mu} \coloneqq \ln\lambda/\lambda = -\ln(-\ln P)/\ln P$ and $\beta \coloneqq 1/\lambda = -1/\ln P$. One can check that the mean $\bar{\mu} + \beta\gamma$ matches μ in Equation 9.9 as $P \to 0$.

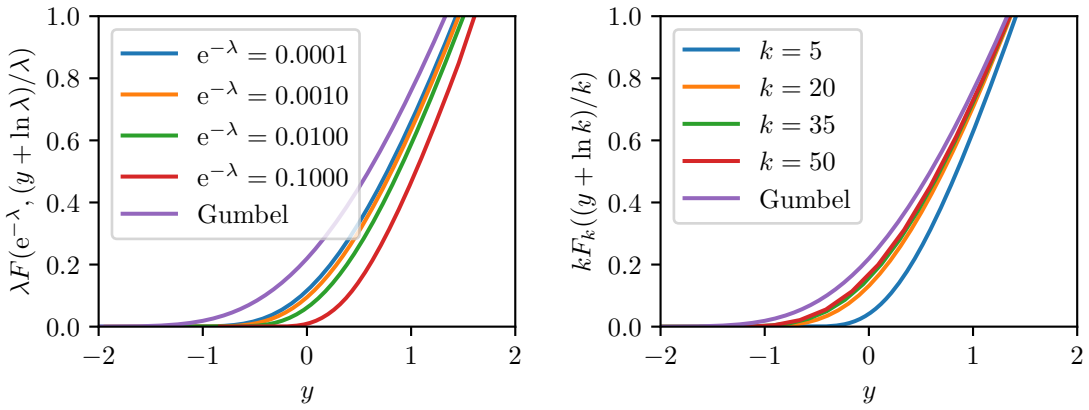


Figure 12.1: The integrated cumulative distribution functions of the continuum limit of the longest run statistics (left) and the longest segment of the broken stick (right) tend to that of the Gumbel distribution.

The relation between the longest run statistics and the Gumbel distribution can be found in literature [7]. Equation 12.1 is indeed closely related to other people's work,

and the only main difference is that the result in the current article gives a continuous probability distribution instead of a discrete one due to us considering the continuum limit.

Another reason why this relation to the Gumbel distribution is interesting is its appearances in higher-dimensional generalizations of the longest run statistics. We may consider the longest run in a Bernoulli sequence as the largest cluster in a 1-dimensional percolation, where a Bernoulli trial is a site that percolates if it is S and does not percolate if it is F. The distribution of the length of the longest run is then the distribution of the largest cluster size in 1-dimensional percolation. The higher-dimensional generalizations of the longest run statistics are then the higher-dimensional percolation models in the subcritical regime (the emphasis on being subcritical can be dropped for the 1-dimensional case because nontrivial 1-dimensional percolation is always subcritical). The exact distribution in the continuum limit is still an open problem for two-dimensional or higher-dimensional percolation, but simulation suggests that the Gumbel distribution is observed [22]. A rigorous proof for the 1-dimensional case serves as a confirmation for the appearance of the Gumbel distribution in percolation.

To prove Equation 12.1, we start by substituting $x = (y + \ln\lambda)/\lambda$ into Equation 9.5 (with μ replaced by the series in Equation 9.7):

$$\lambda F\left(\mathrm{e}^{-\lambda}, \frac{y+\ln\lambda}{\lambda}\right) = -\sum_{s=1}^{\lfloor\lambda/(y+\ln\lambda)\rfloor} \frac{(-1)^s}{s!\,s}\mathrm{e}^{-s(y+\ln\lambda)}\left(\lambda - s\left(y+\ln\lambda\right)\right)^s + \sum_{s=1}^{\infty}\frac{(-1)^s}{s!\,s}\lambda^s + y + \ln\lambda.$$

Combine the second and the fourth term using Equation 9.8 and rearrange the first term:

$$\lambda F\left(\mathrm{e}^{-\lambda}, \frac{y+\ln\lambda}{\lambda}\right) = -\sum_{s=1}^{\infty}\frac{(-1)^s}{s!\,s}\mathrm{e}^{-sy}\left(1 - s\frac{y+\ln\lambda}{\lambda}\right)^{*s} + \mathrm{Ei}(-\lambda) - \gamma + y.$$

In the summand in the first term, for $\lambda \to \infty$ and fixed s, it tends to and is bounded by $\mathrm{e}^{-sy}/s!\,s$, so we can apply the dominated convergence theorem. The term $\mathrm{Ei}(-\lambda)$ tends

to zero. Therefore,

$$\lim_{\lambda\to\infty} \lambda F\left(\mathrm{e}^{-\lambda}, \frac{y+\ln\lambda}{\lambda}\right) = -\sum_{s=1}^{\infty} \frac{(-1)^s}{s!\,s} \mathrm{e}^{-sy} - \gamma + y = -\operatorname{Ei}\left(-\mathrm{e}^{-y}\right),$$

where the last step uses Equation 9.8. This is exactly Equation 12.1, which we wanted to prove.

Furthermore, the Gumbel distribution is also the limit of the broken stick distribution as $k \to \infty$:

$$\lim_{k\to\infty} kF_k\left(\frac{y+\ln k}{k}\right) = G(y)\,. \tag{12.2}$$

This result can also be found in literature [23]. Figure 12.1 plots both sides of Equation 12.2 for various values of k. To prove it, instead of starting from Equation 10.7, we can start from Equation 10.5. Using the dominated convergence theorem, we can prove that

$$\lim_{k\to\infty} \sum_{s=0}^{k+1} \binom{k+1}{s} (-1)^s \left(1 - s\frac{y+\ln k}{k}\right)^{*k} = \exp\left(-\mathrm{e}^{-y}\right) = \partial_y G(y)\,.$$

This is exactly the cumulative distribution function of the standard Gumbel distribution.

The fact that both the continuum limit of the longest run statistics and the broken stick model produce the Gumbel distribution in a certain limit is not coincidental, but it is related to what we discussed in Chapter 11. We can explain it in a handwavy way as the following. From Equation 11.2, we know that $F\left(\mathrm{e}^{-\lambda}, x\right)$ is the sum of $F_k(x)$ over k weighted by the Poisson distribution with parameter λ. When λ is large, the Poisson distribution becomes close to the δ distribution (because its mean λ is much larger than its standard deviation $\sqrt{\lambda}$):

$$\mathrm{e}^{-\lambda}\frac{\lambda^k}{k!} \approx \delta(k-\lambda)\,.$$

We then have

$$F\left(\mathrm{e}^{-\lambda}, x\right) \approx F_\lambda(x)\,,$$

which is why they tend to the same Gumbel distribution.

Chapter 13

Conclusion

This research studies the longest run statistics. We derived a recurrence relation for the longest run distribution. Then, we formulated a continuum limit for the model, and derived the probability density function and the integrated cumulative distribution function in this limit using the Adomian decomposition method. The resultant distribution function is a piecewise analytical function. We showed that this distribution is the exponential generating function of the broken stick distribution, which is the probability distribution of the longest segment of a unit-length interval partitioned at random points. At a certain limit, both distributions tend to the Gumbel distribution, which aligns with findings in researches about percolation models.

Appendix A

Computer codes to reproduce plots

The following Ruby code is used to generate Figure 3.2, Figure 7.2, and Figure 7.3. The method `combo_pc` is also used to generate the coefficients in Table 3.1. To run it, install Ruby, Python, LaTeX, the `matplotlib` Ruby Gem, and the `matplotlib` Python package [24]. The script assumes a Unix-like environment and the `readlink` program from GNU coreutils to find the Python path. If you run in a different environment, adjust the script accordingly.

```
#!/usr/bin/env ruby

ENV['PYTHONPATH'] = Dir.glob(File.expand_path '../../lib/python3.*/site-
    ↪ packages', `readlink -f #{`which python`.strip}`.strip).first

require 'matplotlib/pyplot'

PLT = Matplotlib::Pyplot
PLT.rcParams.update({
  'text.usetex' => true,
  'font.size' => 10,
  'font.family' => 'lmodern',
  'text.latex.preamble' => '\usepackage{lmodern}'
})
```

```
def combo_pc m
  (1..m).each_with_object [[[1]]] do |n, dp|
    dp[n] = Array.new(n+1) { Array.new n+1, 0 }

    # dp[n][0] = (1-Y)**n
    0.upto(n-1) { dp[n][0][_1] = dp[n-1][0][_1] } # will be multiplied by
       ↪ 1-Y later

    1.upto n/2-1 do |k|
      # dp[n][k] = (1-Y) * (Y**k * (0..k).sum { |j| dp[n-k-1][j] } + (0..k
         ↪ -1).sum { |r| Y**r * dp[n-r-1][k] })
      0.upto(k) { |j| 0.upto(n-k-1) { dp[n][k][_1+k] += dp[n-k-1][j][_1] }
         ↪ }
      0.upto(k-1) { |r| 0.upto(n-r-1) { dp[n][k][_1+r] += dp[n-r-1][k][_1]
         ↪ } }
    end

    if n % 2 == 1
      k = n/2
      # dp[n][k] = (1-Y) * (Y**k + (0..k-1).sum { |r| Y**r * dp[n-r-1][k]
         ↪ })
      dp[n][k][k] = 1
      0.upto(k-1) { |r| 0.upto(n-r-1) { dp[n][k][_1+r] += dp[n-r-1][k][_1]
         ↪ } }
    end

    ((n+1)/2).upto n-1 do |k|
      # dp[n][k] = (1-Y) * (Y**k + (0..n-k-1).sum { |r| Y**r * dp[n-r-1][k]
         ↪  })
      dp[n][k][k] = 1
      0.upto(n-k-1) { |r| 0.upto(n-r-1) { dp[n][k][_1+r] += dp[n-r-1][k][_1
         ↪ ] } }
    end

    0.upto(n-1) { |k| n.downto(1) { dp[n][k][_1] -= dp[n][k][_1-1] } } #
       ↪ multiply by 1-Y
```

```
    # dp[n][n] = Y**n
    dp[n][n][n] = 1
  end.last
end

def horner coef, x
  coef.reverse.reduce(0) { |r, c| r * x + c }
end

def inv_coef a
  Array.new a.size do |j|
    binom = nil
    (j..N).sum do |k|
      a[k] * (binom = binom ? binom * k / (k-j) : 1)
    end * (-1)**j
  end
end

N = 50
COEF = combo_pc N
COEF_I = COEF.map { |a| inv_coef a }

def val k, y
  y < 0.5 ? horner(COEF[k], y) : horner(COEF_I[k], 1-y)
end

def plot y, label = '$p=%.3f$'%y
  xs = (0..N).to_a
  ys = xs.map { |k| val k, y }
  PLT.plot xs, ys, label:
end

def plot_finite
  PLT.figure figsize: [4.8, 3.6]
  plot 0.9
  plot 0.92
```

```
  plot 0.94
  plot 0.96
  plot 0.97
  plot 0.98
  plot 0.99
  plot 0.995
  plot 0.999
  PLT.ylim 0, 0.07
  PLT.legend
  PLT.xlabel '$k$'
  PLT.ylabel '$P_{%d,k}(p)$'%N
  PLT.savefig 'n50.pdf', transparent: true, format: :pdf, bbox_inches: :
     ↪ tight
end
plot_finite

include Math

class Integer
  def fac
    (1..self).reduce 1, :*
  end
end

def inf_first y, x
  -y**x*log(y)*(2+log(y**(x-1)))
end

def inf_rest y, x
  smax = x == 0.5 ? 3 : (1/x).ceil
  (2...smax).sum do |s|
    logysx1 = log y**(s*x-1)
    y**(s*x) * logysx1**(s-2) * s*(-1)**s/(s-2).fac * (
      1 + logysx1*2/(s-1) + logysx1**2/s/(s-1)
    )
  end * log(y)
end
```

```
def plot2 y, label = '$P=%.2f$'%y
  xs = Array.new(101) { |i| i/200.0 } + Array.new(101) { |i| 0.5+i/200.0 }
     ↪ + [1]
  ys1 = Array.new 81 do |i|
    x = i/200.0 + 0.1
    inf_first(y,x) + inf_rest(y,x)
  end
  ys2 = Array.new 101 do |i|
    x = i/200.0 + 0.5
    inf_first(y,x)
  end
  ys = [0]*20 + ys1 + ys2 + [5]
  PLT.plot xs, ys, label:
end

def plot_inf
  PLT.figure figsize: [4.8, 3.6]
  plot2 0.01
  plot2 0.03
  plot2 0.09
  plot2 0.3
  plot2 0.6
  plot2 0.9
  PLT.xlabel '$x$'
  PLT.ylabel '$f(P,x)$'
  PLT.ylim 0, 3
  PLT.legend
  PLT.savefig 'inf.pdf', transparent: true, format: :pdf, bbox_inches: :
     ↪ tight
end
plot_inf

def plot3 y
  y **= 1.0/N
  xs = (0..N).map { |k| k/N.to_f }
  ys = (0..N).map { |k| val(k, y)*N }
```

```
  PLT.plot xs, ys
end

def plot_compare
  PLT.figure figsize: [4.8, 3.6]
  plot3 0.01
  plot2 0.01
  plot3 0.2
  plot2 0.2
  plot3 0.5
  plot2 0.5
  plot3 0.9
  plot2 0.9
  PLT.xlabel '$x$'
  PLT.ylabel '$f(P,x)$ or $nP_{N,xn}(P^{1/n})$'
  PLT.ylim 0, 3
  PLT.legend
  PLT.title '$n=%d$'%N
  PLT.savefig 'compare.pdf', transparent: true, format: :pdf, bbox_inches:
     ↪ :tight
end
plot_compare
```

The following Python code is used to generate Figure 9.1, Figure 9.2, Figure 10.1, and Figure 12.1. To run it, install Python, LaTeX, and the `matplotlib` Python package [24].

```
#!/usr/bin/env python

from scipy.special import expi
from numpy import exp, log, linspace, euler_gamma, insert, zeros_like,
   ↪ arange, ceil, where, append
from matplotlib import pyplot as plt

plt.rcParams.update({
  'text.usetex': True,
  'font.size': 10,
```

```
  'font.family': 'lmodern',
  'text.latex.preamble': r'\usepackage{lmodern}'
})

def li(x):
  return expi(log(x))

def repow(x, k):
  return where(x>0, x**k, 0.0)

p = linspace(0.001, 0.999, 999)
mu1 = (li(p) - log(-log(p)) - euler_gamma) / log(p)
p = [0, *p, 1]
mu1 = [0, *mu1, 1]

plt.figure(figsize=(6,2))
plt.plot(p, mu1)
plt.ylim(0, 1)
plt.xlabel('$P$')
plt.ylabel(r'$\mu_1(P)$')
plt.savefig('mu1.pdf', bbox_inches='tight', transparent=True, format='pdf')

def x_and_F(P, upper_x = 1.3):
  lnP = log(P)
  step = 0.005
  hard_zero = 0.02 # F(P,x<hard_zero) is hardcoded to zero
  hard_zero_i = int(hard_zero / step)
  x = arange(0, 1, step)
  if upper_x > 1:
    x = append(x, [upper_x])
  F = zeros_like(x)
  s = 1
  s_fac = 1
  while True:
    x1 = arange(hard_zero, 1/s, step)
    if len(x1) == 0:
      break
```

```
    s_fac *= s
    F[hard_zero_i:hard_zero_i+len(x1)] += P**(s*x1) * ((1-s*x1)*lnP)**s /
       ↪ s_fac / s
    s += 1
  F -= li(P) - log(-lnP) - euler_gamma
  F /= lnP
  F += x
  F[:hard_zero_i] = 0
  return x, F

def plot1(p):
  plt.plot(*x_and_F(p), label='$P=%.2f$'%p)

plt.figure(figsize=(4.8,3.6))
plot1(0.01)
plot1(0.03)
plot1(0.09)
plot1(0.3)
plot1(0.6)
plot1(0.9)
xmin, xmax = plt.xlim()
plt.xlim(xmin, 1.2)
plt.ylim(0, 0.6)
plt.legend()
plt.xlabel('$x$')
plt.ylabel('$F(P,x)$')
plt.savefig('F.pdf', bbox_inches='tight', transparent=True, format='pdf')

def plot_f_k(k):
  if k == 0:
    plt.plot([0, 1, 1], [0, 0, 5], label='$k=0$')
    return
  if k == 1:
    plt.plot([0, 0.5, 0.5, 1], [0, 0, 2, 2], label='$k=1$')
    return
  x = linspace(0, 1, 201)
  f = 0
```

```
  fac = k*(k+1)
  for s in range(1, k+2):
    f += fac * repow(1-s*x, k-1)
    fac *= (k+1-s) / -s
  plt.plot(x, f, label=f'$k={k}$')

plt.figure(figsize=(2.8,2.1))
plot_f_k(0)
plot_f_k(1)
plot_f_k(2)
plot_f_k(3)
plot_f_k(4)
plt.ylim(0, 4)
plt.xlabel('$x$')
plt.ylabel('$f_k(x)$')
plt.savefig('f_k.pdf', bbox_inches='tight', transparent=True, format='pdf')

def x_and_F_k(k, upper_x = 1.3):
  x = linspace(0, 1, 201)
  if upper_x > 1:
    x = append(x, [upper_x])
  F = zeros_like(x)
  fac = 1/(k+1)
  for s in range(1, k+2):
    fac *= (k+2-s) / -s
    F += fac/s * (1 - repow(1-s*x, k+1))
  F += x
  return x, F

def plot_F_k(k):
  plt.plot(*x_and_F_k(k), label=f'$k={k}$')

plt.figure(figsize=(2.8,2.1))
plot_F_k(0)
plot_F_k(1)
plot_F_k(2)
plot_F_k(3)
```

```
plot_F_k(4)
xmin, xmax = plt.xlim()
plt.xlim(xmin, 1.2)
plt.ylim(0, 0.6)
plt.legend()
plt.xlabel('$x$')
plt.ylabel('$F_k(x)$')
plt.savefig('BigF_k.pdf', bbox_inches='tight', transparent=True, format='
    ↪ pdf')

def plot2(P):
  lmd = -log(P)
  x, F = x_and_F(P, upper_x = (2 + log(lmd))/lmd)
  y = x * lmd - log(lmd)
  plt.plot(y, F * lmd, label=r'$\mathrm e^{-\lambda}=%.4f$' % P)

plt.figure(figsize=(2.8,2.1))
plot2(0.0001)
plot2(0.001)
plot2(0.01)
plot2(0.1)
y = linspace(-2, 2, 201)
plt.plot(y, -expi(-exp(-y)), label=f'Gumbel')
plt.xlim(-2, 2)
plt.ylim(0, 1)
plt.legend()
plt.xlabel('$y$')
plt.ylabel(r'$\lambda F(\mathrm e^{-\lambda},(y+\ln\lambda)/\lambda)$')
plt.savefig('Gumbel.pdf', bbox_inches='tight', transparent=True, format='
    ↪ pdf')

def plot_F_k_2(k):
  x, F = x_and_F_k(k, upper_x = (2 + log(k))/k)
  y = x * k - log(k)
  plt.plot(y, F * k, label=r'$k=%i$' % k)

plt.figure(figsize=(2.8,2.1))
```

```
plot_F_k_2(5)
plot_F_k_2(20)
plot_F_k_2(35)
plot_F_k_2(50)
y = linspace(-2, 2, 201)
plt.plot(y, -expi(-exp(-y)), label=f'Gumbel')
plt.xlim(-2, 2)
plt.ylim(0, 1)
plt.legend()
plt.xlabel('$y$')
plt.ylabel(r'$kF_k((y+\ln k)/k)$')
plt.savefig('Gumbel_F_k.pdf', bbox_inches='tight', transparent=True, format
    ↪ ='pdf')
```

Bibliography

[1] N. Balakrishnan and M. V. Koutras, *Runs and scans with applications*. Wiley Series in Probability and Statistics. John Wiley & Sons, Inc, New York, 2002.

[2] A. Wald and J. Wolfowitz, *On a test whether two samples are from the same population*, *The Annals of Mathematical Statistics* **11** (1940), no. 2 147–162.

[3] E. L. De Forest and E. L. DeForrest, *Interpolation and adjustment of series*. Tuttle, Morehouse & Taylor, printers, 1876.

[4] G. J. Hahn and J. B. Gage, *Evaluation of a start-up demonstration test*, *Journal of Quality Technology* **15** (July, 1983) 103–106.

[5] A. N. Philippou, *A note on the fibonacci sequence of order k and the multinomial coefficients*, *The Fibonacci Quarterly* **21** (May, 1983) 82–86.

[6] P. Révész, *Strong theorems on coin tossing*, in *Proceedings of the International Congress of Mathematicians*, vol. 749, Helsinki, 1978.

[7] L. Gordon, M. F. Schilling, and M. S. Waterman, *An extreme value theory for long head runs*, *Probability Theory and Related Fields* **72** (June, 1986) 279–287.

[8] L. Holst, *On the lengths of the pieces of a stick broken at random*, *Journal of Applied Probability* **17** (Sept., 1980) 623–634.

[9] J. E. Cohen, *Species-abundance distributions and taylor's power law of fluctuation scaling*, *Theoretical Ecology* **13** (Dec., 2020) 607–614.

[10] M. Das, T. Bansal, and C. Bhattacharyya, *Ordered stick-breaking prior for sequential mcmc inference of bayesian nonparametric models*, in *Proceedings of the 32nd International Conference on Machine Learning* (F. Bach and D. Blei, eds.), vol. 37 of *Proceedings of Machine Learning Research*, (Lille, France), pp. 550–559, PMLR, 07–09 Jul, 2015.

[11] J. E. Titus-McQuillan, A. V. Nanni, L. M. McIntyre, and R. L. Rogers, *Estimating transcriptome complexities across eukaryotes*, *BMC Genomics* **24** (May, 2023) 254.

[12] K. N. Boyadzhiev, *Exponential polynomials, stirling numbers, and evaluation of some gamma integrals*, *Abstract and Applied Analysis* **2009** (Jan., 2009) 168672.

[13] G. Adomian, *Solving frontier problems of physics: the decomposition method*. Fundamental theories of physics. Springer Science+Business Media, Dordrecht, 1994.

[14] V. Volterra, *Theory of functionals and of integral and integro-differential equations*. Dover New York, 1930.

[15] M. Ernzerhof, *Taylor-series expansion of density functionals*, *Physical Review A* **50** (Dec., 1994) 4593–4607.

[16] G. Hanoch and H. Levy, *The efficiency analysis of choices involving risk*, *The Review of Economic Studies* **36** (July, 1969) 335–346.

[17] M. Abramowitz and I. A. Stegun, *Handbook of mathematical functions: with formulas, graphs, and mathematical tables*. Dover books on mathematics. Dover Publ, New York, NY, 9. dover print.; [nachdr. der ausg. von 1972] ed., 2013.

[18] M. Ceberio, *Uncertainty, Constraints, and Decision Making*. Studies in Systems, Decision and Control Series. Springer, Cham, 1st ed ed., 2023.

[19] I. J. Good, *Some statistical applications of poisson's work*, *Statistical Science* **1** (May, 1986).

[20] P. Flajolet and R. Sedgewick, *Analytic combinatorics*. Cambridge Univ. Press, Cambridge, 4th pr ed., 2013.

[21] E. J. Gumbel, *The return period of flood flows*, *The Annals of Mathematical Statistics* **12** (June, 1941) 163–190.

[22] J. Brzychczyk, *Largest cluster in percolation: Implications for fragmentation studies*, *Physical Review C* **73** (Feb., 2006) 024601.

[23] I. Bairamov, A. Berred, and A. Stepanov, *Limit results for ordered uniform spacings*, *Statistical Papers* **51** (Jan., 2010) 227–240.

[24] J. D. Hunter, *Matplotlib: A 2d graphics environment*, *Computing in Science & Engineering* **9** (2007), no. 3 90–95.